



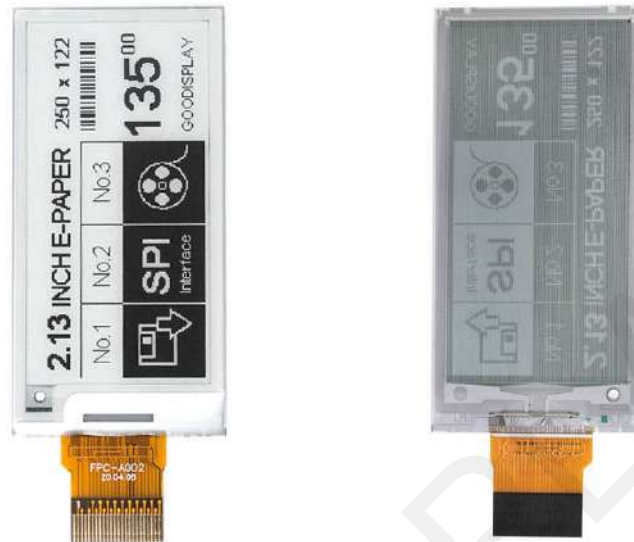
## 2.13 inch E-paper Display Series



**GDEY0213B74**

Dalian Good Display Co., Ltd.

# Product Specifications



|                    |                              |
|--------------------|------------------------------|
| <b>Customer</b>    | <b>Standard</b>              |
| <b>Description</b> | <b>2.13" E-PAPER DISPLAY</b> |
| <b>Model Name</b>  | <b>GDEY0213B74</b>           |
| <b>Date</b>        | <b>2021/03/17</b>            |
| <b>Revision</b>    | <b>1.0</b>                   |

|  | Design Engineering                                                                  |                                                                                       |                                                                                       |
|--|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  | Approval                                                                            | Check                                                                                 | Design                                                                                |
|  |  |  |  |

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**REVISION HISTORY**

| Rev | Date        | Item         | Page | Remark |
|-----|-------------|--------------|------|--------|
| 1.0 | MAR.17.2021 | New Creation | ALL  |        |

GOODDISPLAY

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## 1. Over View

GDEY0213B74 is an Active Matrix Electrophoretic Display (AM EPD), with interface and a reference system design. The display is capable to display images at 1-bit white, black full display capabilities. The 2.13 inch active area contains 250×122 pixels. The module is a TFT-array driving electrophoresis display, with integrated circuits including gate driver, source driver, MCU interface, timing controller, oscillator, DC-DC, SRAM, LUT, VCOM. Module can be used in portable electronic devices, such as Electronic Shelf Label (ESL) System.

## 2. Features

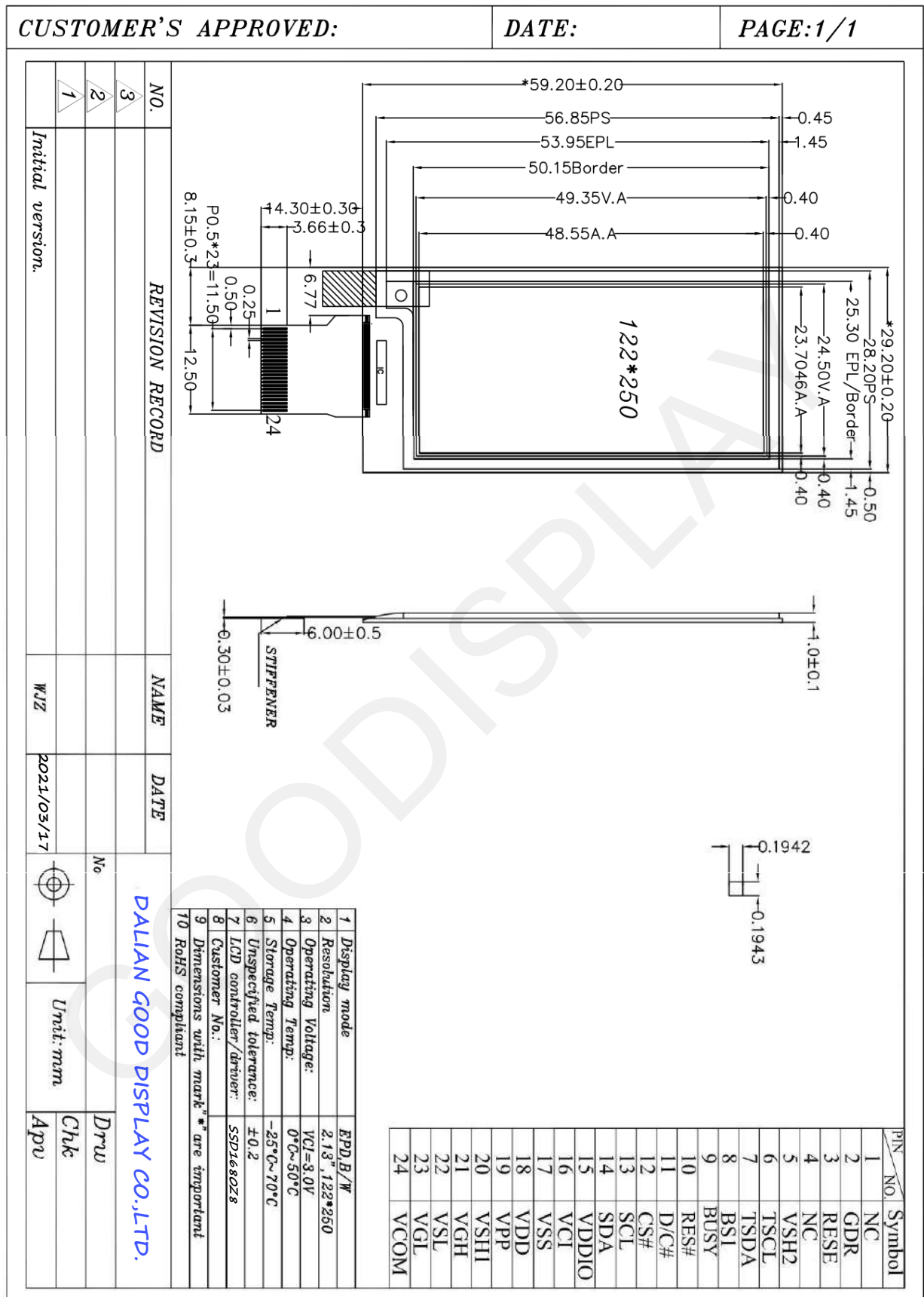
- 250×122 pixels display
- High contrast High reflectance
- Ultra wide viewing angle Ultra low power consumption
- Pure reflective mode
- Bi-stable display
- Commercial temperature range
- Landscape portrait modes
- Hard-coat antiglare display surface
- Ultra Low current deep sleep mode
- On chip display RAM
- Waveform can stored in On-chip OTP or written by MCU
- Serial peripheral interface available
- On-chip oscillator
- On-chip booster and regulator control for generating VCOM, Gate and Source driving voltage
- I2C signal master interface to read external temperature sensor
- Built-in temperature sensor

## 3. Mechanical Specifications

| Parameter           | Specifications           | Unit  | Remark  |
|---------------------|--------------------------|-------|---------|
| Screen Size         | 2.13                     | Inch  |         |
| Display Resolution  | 122(H)×250(V)            | Pixel | Dpi:130 |
| Active Area         | 23.7046×48.55            | mm    |         |
| Pixel Pitch         | 0.1943×0.1942            | mm    |         |
| Pixel Configuration | Square                   |       |         |
| Outline Dimension   | 29.2(H)×59.2 (V) ×1.0(D) | mm    |         |
| Weight              | 3.2±0.5                  | g     |         |



## 4. Mechanical Drawing of EPD module



## 5. Input /Output Pin Assignment

| No. | Name  | I/O | Description                                                                                                        | Remark    |
|-----|-------|-----|--------------------------------------------------------------------------------------------------------------------|-----------|
| 1   | NC    |     | Do not connect with other NC pins                                                                                  | Keep Open |
| 2   | GDR   | O   | N-Channel MOSFET Gate Drive Control                                                                                |           |
| 3   | RESE  | I   | Current Sense Input for the Control Loop                                                                           |           |
| 4   | NC    | NC  | Do not connect with other NC pins                                                                                  | Keep Open |
| 5   | VSH2  | C   | Positive Source driving voltage(Red)                                                                               |           |
| 6   | TSCL  | O   | I <sup>2</sup> C Interface to digital temperature sensor Clock pin                                                 |           |
| 7   | TSDA  | I/O | I <sup>2</sup> C Interface to digital temperature sensor Data pin                                                  |           |
| 8   | BS1   | I   | Bus Interface selection pin                                                                                        | Note 5-5  |
| 9   | BUSY  | O   | Busy state output pin                                                                                              | Note 5-4  |
| 10  | RES#  | I   | Reset signal input. Active Low.                                                                                    | Note 5-3  |
| 11  | D/C#  | I   | Data /Command control pin                                                                                          | Note 5-2  |
| 12  | CS#   | I   | Chip select input pin                                                                                              | Note 5-1  |
| 13  | SCL   | I   | Serial Clock pin (SPI)                                                                                             |           |
| 14  | SDA   | I/O | Serial Data pin (SPI)                                                                                              |           |
| 15  | VDDIO | P   | Power Supply for interface logic pins It should be connected with VCI                                              |           |
| 16  | VCI   | P   | Power Supply for the chip                                                                                          |           |
| 17  | VSS   | P   | Ground                                                                                                             |           |
| 18  | VDD   | C   | Core logic power pin VDD can be regulated internally from VCI. A capacitor should be connected between VDD and VSS |           |
| 19  | VPP   | P   | FOR TEST                                                                                                           |           |
| 20  | VSH1  | C   | Positive Source driving voltage                                                                                    |           |
| 21  | VGH   | C   | Power Supply pin for Positive Gate driving voltage and VSH1                                                        |           |
| 22  | VSL   | C   | Negative Source driving voltage                                                                                    |           |
| 23  | VGL   | C   | Power Supply pin for Negative Gate driving voltage VCOM and VSL                                                    |           |
| 24  | VCOM  | C   | VCOM driving voltage                                                                                               |           |



I = Input Pin, O =Output Pin, I/O = Bi-directional Pin (Input/output), P = Power Pin, C = Capacitor Pin

**Note 5-1:** This pin (CS#) is the chip select input connecting to the MCU. The chip is enabled for MCU communication only when CS# is pulled LOW.

**Note 5-2:** This pin is (D/C#) Data/Command control pin connecting to the MCU in 4-wire SPI mode. When the pin is pulled HIGH, the data at SDA will be interpreted as data. When the pin is pulled LOW, the data at SDA will be interpreted as command.

**Note 5-3:** This pin (RES#) is reset signal input. The Reset is active low.

**Note 5-4:** This pin is Busy state output pin. When Busy is High, the operation of chip should not be interrupted, command should not be sent. The chip would put Busy pin High when -Outputting display waveform -Communicating with digital temperature sensor

**Note 5-5:** Bus interface selection pin

| BS1 State | MCU Interface                                          |
|-----------|--------------------------------------------------------|
| L         | 4-lines serial peripheral interface(SPI) - 8 bits SPI  |
| H         | 3- lines serial peripheral interface(SPI) - 9 bits SPI |

## 6. Electrical Characteristics

### 6.1 Absolute Maximum Rating

| Parameter                | Symbol | Rating           | Unit |
|--------------------------|--------|------------------|------|
| Logic supply voltage     | VCI    | -0.5 to +4.0     | V    |
| Logic Input voltage      | VIN    | -0.5 to VCI +0.5 | V    |
| Logic Output voltage     | VOUT   | -0.5 to VCI +0.5 | V    |
| Operating Temp range     | TOPR   | 0 to +50         | °C   |
| Storage Temp range       | TSTG   | -25 to+70        | °C   |
|                          |        |                  |      |
| Optimal Storage Humidity | HSTGo  | 55±10            | %RH  |

**Note:**

Maximum ratings are those values beyond which damages to the device may occur. Functional operation should be restricted to the limits in the Panel DC Characteristics tables.

### 6.2 Panel DC Characteristics

The following specifications apply for: VSS=0V, VCI=3.0V, TOPR =25°C.

| Parameter                 | Symbol             | Conditions                                                     | Applicable pin | Min.         | Typ.  | Max          | Units   |
|---------------------------|--------------------|----------------------------------------------------------------|----------------|--------------|-------|--------------|---------|
| Single ground             | $V_{SS}$           | -                                                              |                | -            | 0     | -            | V       |
| Logic supply voltage      | $V_{CI}$           | -                                                              | VCI            | 2.2          | 3.0   | 3.7          | V       |
| Core logic voltage        | $V_{DD}$           |                                                                | VDD            | 1.7          | 1.8   | 1.9          | V       |
| High level input voltage  | $V_{IH}$           | -                                                              | -              | 0.8 $V_{CI}$ | -     | -            | V       |
| Low level input voltage   | $V_{IL}$           | -                                                              | -              | -            | -     | 0.2 $V_{CI}$ | V       |
| High level output voltage | $V_{OH}$           | $I_{OH} = -100\mu A$                                           | -              | 0.9 $V_{CI}$ | -     | -            | V       |
| Low level output voltage  | $V_{OL}$           | $I_{OL} = 100\mu A$                                            | -              | -            | -     | 0.1 $V_{CI}$ | V       |
| Typical power             | $P_{TYP}$          | $V_{CI} = 3.0V$                                                | -              | -            | 10.5  | -            | mW      |
| Deep sleep mode           | $P_{STPY}$         | $V_{CI} = 3.0V$                                                | -              | -            | 0.003 | -            | mW      |
| Typical operating current | $I_{opr\_V_{CI}}$  | $V_{CI} = 3.0V$                                                | -              | -            | 3.5   | -            | mA      |
| Full update time          | -                  | 25 °C                                                          | -              | -            | 3     | -            | sec     |
| Fast update time          | -                  | 25 °C                                                          | -              | -            | 1.5   | -            | sec     |
| Partial update time       | -                  | 25 °C                                                          | -              | -            | 0.42  | -            | sec     |
| Sleep mode current        | $I_{slp\_V_{CI}}$  | DC/ DC off<br>No clock<br>No input load<br>Ram data retain     | -              | -            | 20    |              | $\mu A$ |
| Deep sleep mode current   | $I_{dslp\_V_{CI}}$ | DC/ DC off<br>No clock<br>No input load<br>Ram data not retain | -              | -            | 1     | 5            | $\mu A$ |

Notes:

1) Refresh time: the time it takes for the whole process from the screen change to the screen stabilization.

2) The difference between different refresh methods:

Full refresh: The screen will flicker several times during the refresh process;

Fast Refresh: The screen will flash once during the refresh process;

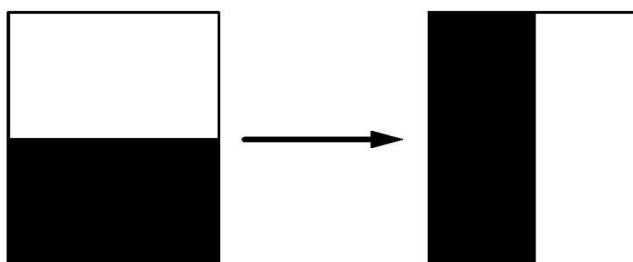
Partial refresh: The screen does not flicker during the refresh process.

During the fast refresh or partial refresh of the electronic paper, it is recommended to add a full-screen refresh after 5 consecutive operations to reduce the accumulation of afterimages on the screen.

1. The typical power is measured with following transition from horizontal 2 scale pattern to vertical 2 scale pattern.

2.The deep sleep power is the consumed power when the panel controller is in deep sleep mode.

3.The listed electrical/optical characteristics are only guaranteed under the controller & waveform provided by Good display.



## 6.3 Panel AC Characteristics

### 6.3.1 MCU Interface Selection

The pin assignment at different interface mode is summarized in Table 6-4-1. Different MCU mode can be set by hardware selection on BS1 pins. The display panel only supports 4-wire SPI or 3-wire SPI interface mode.

| Pin Name         | Data/Command Interface |     | Control Signal |      |      |
|------------------|------------------------|-----|----------------|------|------|
| Bus interface    | SDA                    | SCL | CS#            | D/C# | RES# |
| BS1=L 4-wire SPI | SDA                    | SCL | CS#            | D/C# | RES# |
| BS1=H 3-wire SPI | SDA                    | SCL | CS#            | L    | RES# |

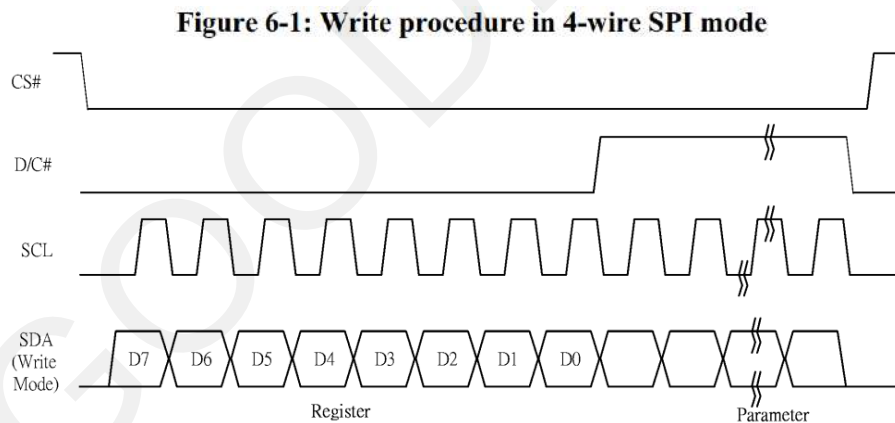
### 6.3.2 MCU Serial Interface (4-wire SPI)

The serial interface consists of serial clock SCL, serial data SDA, D/C#, CS#. This interface supports Write mode and Read mode.

| Function      | CS# | D/C# | SCL |
|---------------|-----|------|-----|
| Write command | L   | L    | ↑   |
| Write data    | L   | H    | ↑   |

**Note:** ↑ stands for rising edge of signal

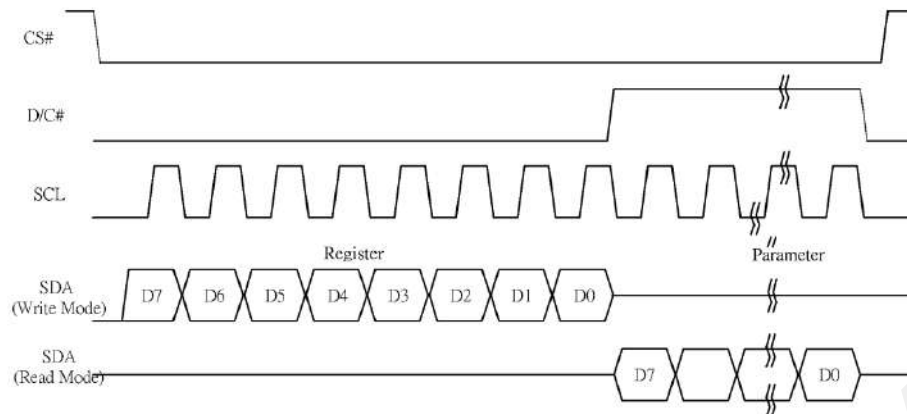
In the write mode SDA is shifted into an 8-bit shift register on every rising edge of SCL in the order of D7, D6, ... D0. The level of D/C# should be kept over the whole byte . The data byte in the shift register is written to the Graphic Display Data RAM /Data Byte register or command Byte register according to D/C# pin.



**In the Read mode:**

1. After driving CS# to low, MCU need to define the register to be read.
2. SDA is shifted into an 8-bit shift register on every rising edge of SCL in the order of D7, D6, ... D0 with D/C# keep low.
3. After SCL change to low for the last bit of register, D/C# need to drive to high.
4. SDA is shifted out an 8-bit data on every falling edge of SCL in the order of D7, D6, ... D0.
5. Depending on register type, more than 1 byte can be read out. After all byte are read, CS# need to drive to high to stop the read operation.

Figure 6-2: Read procedure in 4-wire SPI mode



### 6.3.3 MCU Serial Interface (3-wire SPI)

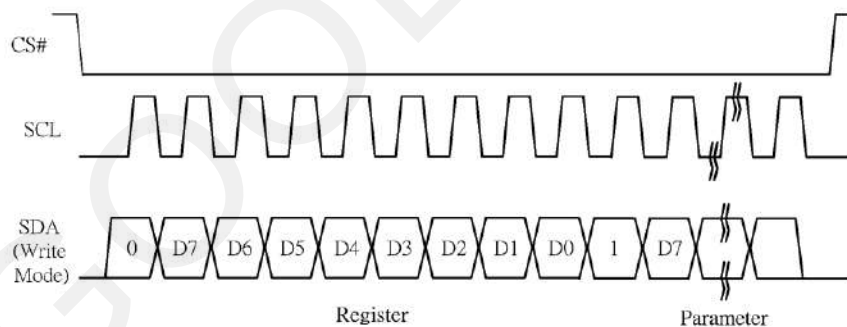
The 3-wire serial interface consists of serial clock SCL, serial data SDA and CS#. This interface also supports Write mode and Read mode.

The operation is similar to 4-wire serial interface while D/C# pin is not used. There are altogether 9-bits will be shifted into the shift register on every ninth clock in sequence: D/C# bit, D7 to D0 bit. The D/C# bit (first bit of the sequential data) will determine the following data byte in the shift register is written to the Display Data RAM (D/C# bit = 1) or the command register (D/C# bit = 0).

| Function      | CS# | D/C# | SCL |
|---------------|-----|------|-----|
| Write command | L   | Tie  | ↑   |
| Write data    | L   | Tie  | ↑   |

**Note:** ↑ stands for rising edge of signal

Figure 6-3: Write procedure in 3-wire SPI mode

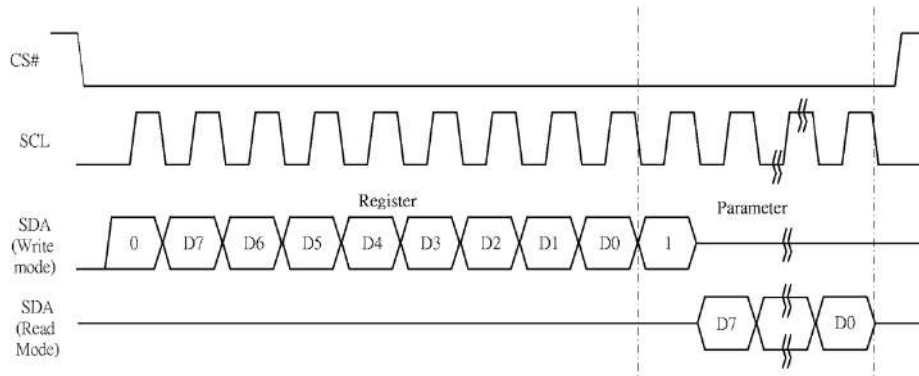


In the Read mode:

1. After driving CS# to low, MCU need to define the register to be read.
2. D/C=0 is shifted thru SDA with one rising edge of SCL
3. SDA is shifted into an 8-bit shift register on every rising edge of SCL in the order of D7, D6, ... D0.
4. D/C=1 is shifted thru SDA with one rising edge of SCL
5. SDA is shifted out an 8-bit data on every falling edge of SCL in the order of D7, D6, ... D0.
6. Depending on register type, more than 1 byte can be read out. After all byte are read, CS# need to drive to high to stop the read operation.

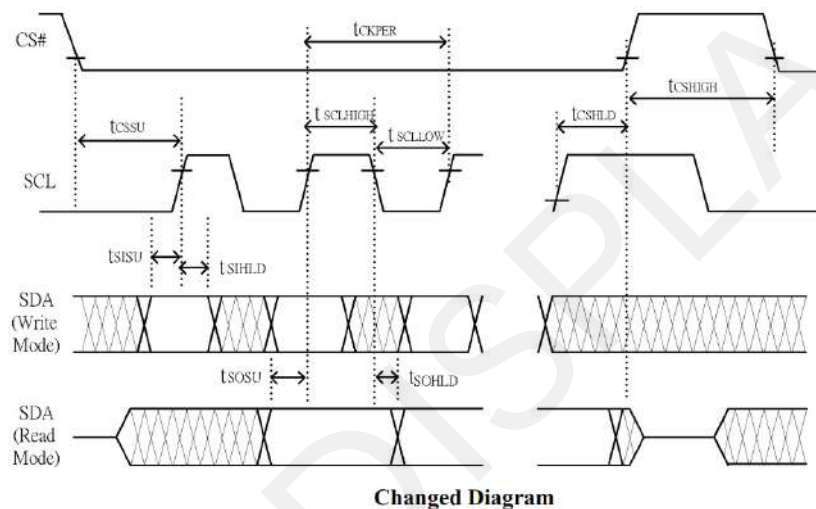


**Figure 6-4: Read procedure in 3-wire SPI mode**



### 6.3.4 Interface Timing

The following specifications apply for: VSS=0V, VCI=3.0V, TOPR =25°C.



#### Serial Interface Timing Characteristics

(VCI - VSS = 2.2V to 3.7V, TOPR = 25°C, CL=20pF)

##### Write mode

| Symbol        | Parameter                                                                    | Min | Typ | Max | Unit |
|---------------|------------------------------------------------------------------------------|-----|-----|-----|------|
| $f_{SCL}$     | SCL frequency (Write Mode)                                                   |     |     | 20  | MHz  |
| $t_{CSSU}$    | Time CS# has to be low before the first rising edge of SCLK                  | 60  |     |     | ns   |
| $t_{CSSL}$    | Time CS# has to remain low after the last falling edge of SCLK               | 65  |     |     | ns   |
| $t_{CSHIGH}$  | Time CS# has to remain high between two transfers                            | 100 |     |     | ns   |
| $t_{SCLHIGH}$ | Part of the clock period where SCL has to remain high                        | 25  |     |     | ns   |
| $t_{SCLLOW}$  | Part of the clock period where SCL has to remain low                         | 25  |     |     | ns   |
| $t_{SI}$      | Time SI (SDA Write Mode) has to be stable before the next rising edge of SCL | 10  |     |     | ns   |
| $t_{SHLD}$    | Time SI (SDA Write Mode) has to remain stable after the rising edge of SCL   | 40  |     |     | ns   |

##### Read mode

| Symbol        | Parameter                                                                | Min | Typ | Max | Unit |
|---------------|--------------------------------------------------------------------------|-----|-----|-----|------|
| $f_{SCL}$     | SCL frequency (Read Mode)                                                |     |     | 2.5 | MHz  |
| $t_{CSSU}$    | Time CS# has to be low before the first rising edge of SCLK              | 100 |     |     | ns   |
| $t_{CSSL}$    | Time CS# has to remain low after the last falling edge of SCLK           | 50  |     |     | ns   |
| $t_{CSHIGH}$  | Time CS# has to remain high between two transfers                        | 250 |     |     | ns   |
| $t_{SCLHIGH}$ | Part of the clock period where SCL has to remain high                    | 180 |     |     | ns   |
| $t_{SCLLOW}$  | Part of the clock period where SCL has to remain low                     | 180 |     |     | ns   |
| $t_{SOSU}$    | Time SO(SDA Read Mode) will be stable before the next rising edge of SCL |     | 50  |     | ns   |
| $t_{SOHLD}$   | Time SO (SDA Read Mode) will remain stable after the falling edge of SCL |     | 0   |     | ns   |

## 7. Command Table

| Command Table |      |        |                |                |                |                |                |                |                |                | Command                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |     |        |     |     |    |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |       |    |     |      |  |  |
|---------------|------|--------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-----|--------|-----|-----|----|-----|----|-----|----|-----|------|-----|------|-----|----|-----|----|-----|------|-----|------|-----|----|-----|----|-----|------|-----|------|-----|----|-----|----|-----|------|-----|------|-----|----|-----|----|-----|------|-----|------|-----|----|-----|----|-------|----|-----|------|--|--|
| R/W#          | D/C# | Hex    | D7             | D6             | D5             | D4             | D3             | D2             | D1             | D0             |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |     |        |     |     |    |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |       |    |     |      |  |  |
| 0             | 0    | 01     | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 1              | Driver Output control        | Gate setting<br>A[8:0]= 127h [POR], 296 MUX<br>MUX Gate lines setting as (A[8:0] + 1).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |     |        |     |     |    |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |       |    |     |      |  |  |
| 0             | 1    |        | A <sub>7</sub> | A <sub>6</sub> | A <sub>5</sub> | A <sub>4</sub> | A <sub>3</sub> | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |     |        |     |     |    |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |       |    |     |      |  |  |
| 0             | 1    |        | 0              | 0              | 0              | 0              | 0              | 0              | 0              | A <sub>8</sub> |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |     |        |     |     |    |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |       |    |     |      |  |  |
| 0             | 1    |        | 0              | 0              | 0              | 0              | 0              | B <sub>2</sub> | B <sub>1</sub> | B <sub>0</sub> |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |     |        |     |     |    |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |       |    |     |      |  |  |
|               |      |        |                |                |                |                |                |                |                |                |                              | B[2:0] = 000 [POR].<br>Gate scanning sequence and direction<br><br>B[2]: GD<br>Selects the 1st output Gate<br>GD=0 [POR],<br>G0 is the 1st gate output channel, gate output sequence is G0,G1, G2, G3, ...<br>GD=1,<br>G1 is the 1st gate output channel, gate output sequence is G1, G0, G3, G2, ...<br><br>B[1]: SM<br>Change scanning order of gate driver.<br>SM=0 [POR],<br>G0, G1, G2, G3...295 (left and right gate interlaced)<br>SM=1,<br>G0, G2, G4 ...G294, G1, G3, ...G295<br><br>B[0]: TB<br>TB = 0 [POR], scan from G0 to G295<br>TB = 1, scan from G295 to G0.                                                                                                                                                                                                                                                                |        |     |        |     |     |    |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |       |    |     |      |  |  |
| 0             | 0    | 03     | 0              | 0              | 0              | 0              | 0              | 0              | 1              | 1              | Gate Driving voltage Control | Set Gate driving voltage<br>A[4:0] = 00h [POR]<br>VGH setting from 10V to 20V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |     |        |     |     |    |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |       |    |     |      |  |  |
| 0             | 1    |        | 0              | 0              | 0              | A <sub>4</sub> | A <sub>3</sub> | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |     |        |     |     |    |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |       |    |     |      |  |  |
|               |      |        |                |                |                |                |                |                |                |                |                              | <table><tr><th>A[4:0]</th><th>VGH</th><th>A[4:0]</th><th>VGH</th></tr><tr><td>00h</td><td>20</td><td>0Dh</td><td>15</td></tr><tr><td>03h</td><td>10</td><td>0Eh</td><td>15.5</td></tr><tr><td>04h</td><td>10.5</td><td>0Fh</td><td>16</td></tr><tr><td>05h</td><td>11</td><td>10h</td><td>16.5</td></tr><tr><td>06h</td><td>11.5</td><td>11h</td><td>17</td></tr><tr><td>07h</td><td>12</td><td>12h</td><td>17.5</td></tr><tr><td>08h</td><td>12.5</td><td>13h</td><td>18</td></tr><tr><td>07h</td><td>12</td><td>14h</td><td>18.5</td></tr><tr><td>08h</td><td>12.5</td><td>15h</td><td>19</td></tr><tr><td>09h</td><td>13</td><td>16h</td><td>19.5</td></tr><tr><td>0Ah</td><td>13.5</td><td>17h</td><td>20</td></tr><tr><td>0Bh</td><td>14</td><td>Other</td><td>NA</td></tr><tr><td>0Ch</td><td>14.5</td><td></td><td></td></tr></table> | A[4:0] | VGH | A[4:0] | VGH | 00h | 20 | 0Dh | 15 | 03h | 10 | 0Eh | 15.5 | 04h | 10.5 | 0Fh | 16 | 05h | 11 | 10h | 16.5 | 06h | 11.5 | 11h | 17 | 07h | 12 | 12h | 17.5 | 08h | 12.5 | 13h | 18 | 07h | 12 | 14h | 18.5 | 08h | 12.5 | 15h | 19 | 09h | 13 | 16h | 19.5 | 0Ah | 13.5 | 17h | 20 | 0Bh | 14 | Other | NA | 0Ch | 14.5 |  |  |
| A[4:0]        | VGH  | A[4:0] | VGH            |                |                |                |                |                |                |                |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |     |        |     |     |    |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |       |    |     |      |  |  |
| 00h           | 20   | 0Dh    | 15             |                |                |                |                |                |                |                |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |     |        |     |     |    |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |       |    |     |      |  |  |
| 03h           | 10   | 0Eh    | 15.5           |                |                |                |                |                |                |                |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |     |        |     |     |    |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |       |    |     |      |  |  |
| 04h           | 10.5 | 0Fh    | 16             |                |                |                |                |                |                |                |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |     |        |     |     |    |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |       |    |     |      |  |  |
| 05h           | 11   | 10h    | 16.5           |                |                |                |                |                |                |                |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |     |        |     |     |    |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |       |    |     |      |  |  |
| 06h           | 11.5 | 11h    | 17             |                |                |                |                |                |                |                |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |     |        |     |     |    |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |       |    |     |      |  |  |
| 07h           | 12   | 12h    | 17.5           |                |                |                |                |                |                |                |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |     |        |     |     |    |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |       |    |     |      |  |  |
| 08h           | 12.5 | 13h    | 18             |                |                |                |                |                |                |                |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |     |        |     |     |    |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |       |    |     |      |  |  |
| 07h           | 12   | 14h    | 18.5           |                |                |                |                |                |                |                |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |     |        |     |     |    |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |       |    |     |      |  |  |
| 08h           | 12.5 | 15h    | 19             |                |                |                |                |                |                |                |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |     |        |     |     |    |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |       |    |     |      |  |  |
| 09h           | 13   | 16h    | 19.5           |                |                |                |                |                |                |                |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |     |        |     |     |    |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |       |    |     |      |  |  |
| 0Ah           | 13.5 | 17h    | 20             |                |                |                |                |                |                |                |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |     |        |     |     |    |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |       |    |     |      |  |  |
| 0Bh           | 14   | Other  | NA             |                |                |                |                |                |                |                |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |     |        |     |     |    |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |       |    |     |      |  |  |
| 0Ch           | 14.5 |        |                |                |                |                |                |                |                |                |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |     |        |     |     |    |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |     |      |     |      |     |    |     |    |       |    |     |      |  |  |



**Command Table**

| R/W#                                                          | D/C# | Hex       | D7             | D6             | D5             | D4             | D3             | D2             | D1             | D0             | Command                        | Description                                                                                                                                               |                                                            |                                           |  |
|---------------------------------------------------------------|------|-----------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|-------------------------------------------|--|
| 0                                                             | 0    | 04        | 0              | 0              | 0              | 0              | 0              | 1              | 0              | 0              | Source Driving voltage Control | Set Source driving voltage<br>A[7:0] = 41h [POR], VSH1 at 15V<br>B[7:0] = A8h [POR], VSH2 at 5V.<br>C[7:0] = 32h [POR], VSL at -15V<br>Remark: VSH1>=VSH2 |                                                            |                                           |  |
| 0                                                             | 1    |           | A <sub>7</sub> | A <sub>6</sub> | A <sub>5</sub> | A <sub>4</sub> | A <sub>3</sub> | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> |                                |                                                                                                                                                           |                                                            |                                           |  |
| 0                                                             | 1    |           | B <sub>7</sub> | B <sub>6</sub> | B <sub>5</sub> | B <sub>4</sub> | B <sub>3</sub> | B <sub>2</sub> | B <sub>1</sub> | B <sub>0</sub> |                                |                                                                                                                                                           |                                                            |                                           |  |
| 0                                                             | 1    |           | C <sub>7</sub> | C <sub>6</sub> | C <sub>5</sub> | C <sub>4</sub> | C <sub>3</sub> | C <sub>2</sub> | C <sub>1</sub> | C <sub>0</sub> |                                |                                                                                                                                                           |                                                            |                                           |  |
| A[7]/B[7] = 1,<br>VSH1/VSH2 voltage setting from 2.4V to 8.8V |      |           |                |                |                |                |                |                |                |                |                                |                                                                                                                                                           | A[7]/B[7] = 0,<br>VSH1/VSH2 voltage setting from 9V to 17V | C[7] = 0,<br>VSL setting from -5V to -17V |  |
| A/B[7:0]                                                      |      | VSH1/VSH2 |                | A/B[7:0]       |                | VSH1/VSH2      |                | A/B[7:0]       |                | VSH1/VSH2      |                                | C[7:0]                                                                                                                                                    | VSL                                                        |                                           |  |
| 8Eh                                                           |      | 2.4       |                | AFh            |                | 5.7            |                | 23h            |                | 9              |                                | 3Ch                                                                                                                                                       |                                                            | 14                                        |  |
| 8Fh                                                           |      | 2.5       |                | B0h            |                | 5.8            |                | 24h            |                | 9.2            |                                | 3Dh                                                                                                                                                       |                                                            | 14.2                                      |  |
| 90h                                                           |      | 2.6       |                | B1h            |                | 5.9            |                | 25h            |                | 9.4            |                                | 3Eh                                                                                                                                                       |                                                            | 14.4                                      |  |
| 91h                                                           |      | 2.7       |                | B2h            |                | 6              |                | 26h            |                | 9.6            |                                | 3Fh                                                                                                                                                       |                                                            | 14.6                                      |  |
| 92h                                                           |      | 2.8       |                | B3h            |                | 6.1            |                | 27h            |                | 9.8            |                                | 40h                                                                                                                                                       |                                                            | 14.8                                      |  |
| 93h                                                           |      | 2.9       |                | B4h            |                | 6.2            |                | 28h            |                | 10             |                                | 41h                                                                                                                                                       |                                                            | 15                                        |  |
| 94h                                                           |      | 3         |                | B5h            |                | 6.3            |                | 29h            |                | 10.2           |                                | 42h                                                                                                                                                       |                                                            | 15.2                                      |  |
| 95h                                                           |      | 3.1       |                | B6h            |                | 6.4            |                | 2Ah            |                | 10.4           |                                | 43h                                                                                                                                                       |                                                            | 15.4                                      |  |
| 96h                                                           |      | 3.2       |                | B7h            |                | 6.5            |                | 2Bh            |                | 10.6           |                                | 44h                                                                                                                                                       |                                                            | 15.6                                      |  |
| 97h                                                           |      | 3.3       |                | B8h            |                | 6.6            |                | 2Ch            |                | 10.8           |                                | 45h                                                                                                                                                       |                                                            | 15.8                                      |  |
| 98h                                                           |      | 3.4       |                | B9h            |                | 6.7            |                | 2Dh            |                | 11             |                                | 46h                                                                                                                                                       |                                                            | 16                                        |  |
| 99h                                                           |      | 3.5       |                | BAh            |                | 6.8            |                | 2Eh            |                | 11.2           |                                | 47h                                                                                                                                                       |                                                            | 16.2                                      |  |
| 9Ah                                                           |      | 3.6       |                | BBh            |                | 6.9            |                | 2Fh            |                | 11.4           |                                | 48h                                                                                                                                                       |                                                            | 16.4                                      |  |
| 9Bh                                                           |      | 3.7       |                | BCh            |                | 7              |                | 30h            |                | 11.6           |                                | 49h                                                                                                                                                       |                                                            | 16.6                                      |  |
| 9Ch                                                           |      | 3.8       |                | BDh            |                | 7.1            |                | 31h            |                | 11.8           |                                | 4Ah                                                                                                                                                       |                                                            | 16.8                                      |  |
| 9Dh                                                           |      | 3.9       |                | BEh            |                | 7.2            |                | 32h            |                | 12             |                                | 4Bh                                                                                                                                                       |                                                            | 17                                        |  |
| 9Eh                                                           |      | 4         |                | BFh            |                | 7.3            |                | 33h            |                | 12.2           |                                | Other                                                                                                                                                     |                                                            | NA                                        |  |
| 9Fh                                                           |      | 4.1       |                | C0h            |                | 7.4            |                | 34h            |                | 12.4           |                                |                                                                                                                                                           |                                                            |                                           |  |
| A0h                                                           |      | 4.2       |                | C1h            |                | 7.5            |                | 35h            |                | 12.6           |                                |                                                                                                                                                           |                                                            |                                           |  |
| A1h                                                           |      | 4.3       |                | C2h            |                | 7.6            |                | 36h            |                | 12.8           |                                |                                                                                                                                                           |                                                            |                                           |  |
| A2h                                                           |      | 4.4       |                | C3h            |                | 7.7            |                | 37h            |                | 13             |                                |                                                                                                                                                           |                                                            |                                           |  |
| A3h                                                           |      | 4.5       |                | C4h            |                | 7.8            |                | 38h            |                | 13.2           |                                |                                                                                                                                                           |                                                            |                                           |  |
| A4h                                                           |      | 4.6       |                | C5h            |                | 7.9            |                | 39h            |                | 13.4           |                                |                                                                                                                                                           |                                                            |                                           |  |
| A5h                                                           |      | 4.7       |                | C6h            |                | 8              |                | 3Ah            |                | 13.6           |                                |                                                                                                                                                           |                                                            |                                           |  |
| A6h                                                           |      | 4.8       |                | C7h            |                | 8.1            |                | 3Bh            |                | 13.8           |                                |                                                                                                                                                           |                                                            |                                           |  |
| A7h                                                           |      | 4.9       |                | C8h            |                | 8.2            |                |                |                |                |                                |                                                                                                                                                           |                                                            |                                           |  |
| A8h                                                           |      | 5         |                | C9h            |                | 8.3            |                |                |                |                |                                |                                                                                                                                                           |                                                            |                                           |  |
| A9h                                                           |      | 5.1       |                | CAh            |                | 8.4            |                |                |                |                |                                |                                                                                                                                                           |                                                            |                                           |  |
| AAh                                                           |      | 5.2       |                | CBh            |                | 8.5            |                |                |                |                |                                |                                                                                                                                                           |                                                            |                                           |  |
| ABh                                                           |      | 5.3       |                | CCh            |                | 8.6            |                |                |                |                |                                |                                                                                                                                                           |                                                            |                                           |  |
| ACh                                                           |      | 5.4       |                | CDh            |                | 8.7            |                |                |                |                |                                |                                                                                                                                                           |                                                            |                                           |  |
| ADh                                                           |      | 5.5       |                | CEh            |                | 8.8            |                |                |                |                |                                |                                                                                                                                                           |                                                            |                                           |  |
| AEh                                                           |      | 5.6       |                | Other          |                | NA             |                |                |                |                |                                |                                                                                                                                                           |                                                            |                                           |  |

|   |   |    |   |   |   |   |   |   |   |   |                                     |                                                                                                                                                           |
|---|---|----|---|---|---|---|---|---|---|---|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0 | 0 | 08 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | Initial Code Setting<br>OTP Program | Program Initial Code Setting<br><br>The command required CLKEN=1.<br>Refer to Register 0x22 for detail.<br>BUSY pad will output high during<br>operation. |
|---|---|----|---|---|---|---|---|---|---|---|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|

|   |   |    |                |                |                |                |                |                |                |                |                                            |                                                                                                                                                    |
|---|---|----|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 0 | 0 | 09 | 0              | 0              | 0              | 0              | 1              | 0              | 0              | 1              | Write Register for Initial<br>Code Setting | Write Register for Initial Code Setting<br>Selection<br>A[7:0] ~ D[7:0]: Reserved<br>Details refer to Application Notes of Initial<br>Code Setting |
| 0 | 1 |    | A <sub>7</sub> | A <sub>6</sub> | A <sub>5</sub> | A <sub>4</sub> | A <sub>3</sub> | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> |                                            |                                                                                                                                                    |
| 0 | 1 |    | B <sub>7</sub> | B <sub>6</sub> | B <sub>5</sub> | B <sub>4</sub> | B <sub>3</sub> | B <sub>2</sub> | B <sub>1</sub> | B <sub>0</sub> |                                            |                                                                                                                                                    |
| 0 | 1 |    | C <sub>7</sub> | C <sub>6</sub> | C <sub>5</sub> | C <sub>4</sub> | C <sub>3</sub> | C <sub>2</sub> | C <sub>1</sub> | C <sub>0</sub> |                                            |                                                                                                                                                    |
| 0 | 1 |    | D <sub>7</sub> | D <sub>6</sub> | D <sub>5</sub> | D <sub>4</sub> | D <sub>3</sub> | D <sub>2</sub> | D <sub>1</sub> | D <sub>0</sub> |                                            |                                                                                                                                                    |

|   |   |    |   |   |   |   |   |   |   |   |                                           |                                        |
|---|---|----|---|---|---|---|---|---|---|---|-------------------------------------------|----------------------------------------|
| 0 | 0 | 0A | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | Read Register for Initial<br>Code Setting | Read Register for Initial Code Setting |
|---|---|----|---|---|---|---|---|---|---|---|-------------------------------------------|----------------------------------------|

| Command Table |      |     |    |                |                |                |                |                |                |                |                            |                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------|------|-----|----|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R/W#          | D/C# | Hex | D7 | D6             | D5             | D4             | D3             | D2             | D1             | D0             | Command                    | Description                                                                                                                                                                                                                                                                                                                                                                                    |
| 0             | 0    | 0C  | 0  | 0              | 0              | 0              | 1              | 1              | 0              | 0              | Booster Soft start Control | Booster Enable with Phase 1, Phase 2 and Phase 3 for soft start current and duration setting.<br><br>A[7:0] -> Soft start setting for Phase1<br>= 8Bh [POR]<br>B[7:0] -> Soft start setting for Phase2<br>= 9Ch [POR]<br>C[7:0] -> Soft start setting for Phase3<br>= 96h [POR]<br>D[7:0] -> Duration setting<br>= 0Fh [POR]<br><br>Bit Description of each byte:<br>A[6:0] / B[6:0] / C[6:0]: |
| 0             | 1    |     | 1  | A <sub>6</sub> | A <sub>5</sub> | A <sub>4</sub> | A <sub>3</sub> | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> |                            |                                                                                                                                                                                                                                                                                                                                                                                                |
| 0             | 1    |     | 1  | B <sub>6</sub> | B <sub>5</sub> | B <sub>4</sub> | B <sub>3</sub> | B <sub>2</sub> | B <sub>1</sub> | B <sub>0</sub> |                            |                                                                                                                                                                                                                                                                                                                                                                                                |
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|   |   |    |   |                |                |                |   |                |                |                |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---|---|----|---|----------------|----------------|----------------|---|----------------|----------------|----------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0 | 0 | 11 | 0 | 0              | 0              | 1              | 0 | 0              | 0              | 1              | Data Entry mode setting | <p>Define data entry sequence<br/>A[2:0] = 011 [POR]</p> <p>A [1:0] = ID[1:0]<br/>Address automatic increment / decrement setting<br/>The setting of incrementing or decrementing of the address counter can be made independently in each upper and lower bit of the address.<br/>00 –Y decrement, X decrement,<br/>01 –Y decrement, X increment,<br/>10 –Y increment, X decrement,<br/>11 –Y increment, X increment [POR]</p> <p>A[2] = AM<br/>Set the direction in which the address counter is updated automatically after data are written to the RAM.<br/>AM= 0, the address counter is updated in the X direction. [POR]<br/>AM = 1, the address counter is updated in the Y direction.</p> |
| 0 | 1 |    | 0 | 0              | 0              | 0              | 0 | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 0 | 0 | 12 | 0 | 0              | 0              | 1              | 0 | 0              | 1              | 0              | SW RESET                | <p>It resets the commands and parameters to their S/W Reset default values except R10h-Deep Sleep Mode</p> <p>During operation, BUSY pad will output high.</p> <p>Note: RAM are unaffected by this command.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 0 | 0 | 14 | 0 | 0              | 0              | 1              | 0 | 1              | 0              | 0              | HV Ready Detection      | <p>HV ready detection<br/>A[7:0] = 00h [POR]<br/>The command required CLKEN=1 and ANALOGEN=1.<br/>Refer to Register 0x22 for detail.<br/>After this command initiated, HV Ready detection starts.<br/>BUSY pad will output high during detection.<br/>The detection result can be read from the Status Bit Read (Command 0x2F).</p>                                                                                                                                                                                                                                                                                                                                                                |
| 0 | 1 |    | 0 | A <sub>6</sub> | A <sub>5</sub> | A <sub>4</sub> | 0 | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> |                         | <p>A[6:4]=n for cool down duration:<br/>10ms x (n+1)<br/>A[2:0]=m for number of Cool Down Loop to detect.<br/>The max HV ready duration is<br/>10ms x (n+1) x (m)<br/>HV ready detection will be trigger after each cool down time. The detection will be completed when HV is ready.<br/>For 1 shot HV ready detection, A[7:0] can be set as 00h.</p>                                                                                                                                                                                                                                                                                                                                             |



|        |                                                 |    |                 |                 |                |                |                |                |                |                |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                              |     |                   |     |                                   |     |                                                 |     |         |     |      |       |    |
|--------|-------------------------------------------------|----|-----------------|-----------------|----------------|----------------|----------------|----------------|----------------|----------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------------------------------|-----|-------------------|-----|-----------------------------------|-----|-------------------------------------------------|-----|---------|-----|------|-------|----|
| 0      | 0                                               | 15 | 0               | 0               | 0              | 1              | 0              | 1              | 0              | 1              | VCI Detection                                                             | VCI Detection<br>A[2:0] = 100 [POR] , Detect level at 2.3V<br>A[2:0] : VCI level Detect <table><tr><td>A[2:0]</td><td>VCI level</td></tr><tr><td>011</td><td>2.2V</td></tr><tr><td>100</td><td>2.3V</td></tr><tr><td>101</td><td>2.4V</td></tr><tr><td>110</td><td>2.5V</td></tr><tr><td>111</td><td>2.6V</td></tr><tr><td>Other</td><td>NA</td></tr></table><br>The command required CLKEN=1 and ANALOGEN=1<br>Refer to Register 0x22 for detail.<br><br>After this command initiated, VCI detection starts.<br>BUSY pad will output high during detection.<br>The detection result can be read from the Status Bit Read (Command 0x2F).                                                                                                                      | A[2:0] | VCI level                    | 011 | 2.2V              | 100 | 2.3V                              | 101 | 2.4V                                            | 110 | 2.5V    | 111 | 2.6V | Other | NA |
| A[2:0] | VCI level                                       |    |                 |                 |                |                |                |                |                |                |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                              |     |                   |     |                                   |     |                                                 |     |         |     |      |       |    |
| 011    | 2.2V                                            |    |                 |                 |                |                |                |                |                |                |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                              |     |                   |     |                                   |     |                                                 |     |         |     |      |       |    |
| 100    | 2.3V                                            |    |                 |                 |                |                |                |                |                |                |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                              |     |                   |     |                                   |     |                                                 |     |         |     |      |       |    |
| 101    | 2.4V                                            |    |                 |                 |                |                |                |                |                |                |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                              |     |                   |     |                                   |     |                                                 |     |         |     |      |       |    |
| 110    | 2.5V                                            |    |                 |                 |                |                |                |                |                |                |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                              |     |                   |     |                                   |     |                                                 |     |         |     |      |       |    |
| 111    | 2.6V                                            |    |                 |                 |                |                |                |                |                |                |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                              |     |                   |     |                                   |     |                                                 |     |         |     |      |       |    |
| Other  | NA                                              |    |                 |                 |                |                |                |                |                |                |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                              |     |                   |     |                                   |     |                                                 |     |         |     |      |       |    |
| 0      | 1                                               |    | 0               | 0               | 0              | 0              | 0              | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                              |     |                   |     |                                   |     |                                                 |     |         |     |      |       |    |
| 0      | 0                                               | 18 | 0               | 0               | 0              | 1              | 1              | 0              | 0              | 0              | Temperature Sensor Control                                                | Temperature Sensor Selection<br>A[7:0] = 48h [POR], external temperature sensor<br>A[7:0] = 80h Internal temperature sensor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |                              |     |                   |     |                                   |     |                                                 |     |         |     |      |       |    |
| 0      | 1                                               |    | A <sub>7</sub>  | A <sub>6</sub>  | A <sub>5</sub> | A <sub>4</sub> | A <sub>3</sub> | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                              |     |                   |     |                                   |     |                                                 |     |         |     |      |       |    |
| 0      | 0                                               | 1A | 0               | 0               | 0              | 1              | 1              | 0              | 1              | 0              | Temperature Sensor Control (Write to temperature register)                | Write to temperature register.<br>A[11:0] = 7FFh [POR]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |                              |     |                   |     |                                   |     |                                                 |     |         |     |      |       |    |
| 0      | 1                                               |    | A <sub>11</sub> | A <sub>10</sub> | A <sub>9</sub> | A <sub>8</sub> | A <sub>7</sub> | A <sub>6</sub> | A <sub>5</sub> | A <sub>4</sub> |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                              |     |                   |     |                                   |     |                                                 |     |         |     |      |       |    |
| 0      | 1                                               |    | A <sub>3</sub>  | A <sub>2</sub>  | A <sub>1</sub> | A <sub>0</sub> | 0              | 0              | 0              | 0              |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                              |     |                   |     |                                   |     |                                                 |     |         |     |      |       |    |
| 0      | 0                                               | 1B | 0               | 0               | 0              | 1              | 1              | 0              | 1              | 1              | Temperature Sensor Control (Read from temperature register)               | Read from temperature register.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                              |     |                   |     |                                   |     |                                                 |     |         |     |      |       |    |
| 1      | 1                                               |    | A <sub>11</sub> | A <sub>10</sub> | A <sub>9</sub> | A <sub>8</sub> | A <sub>7</sub> | A <sub>6</sub> | A <sub>5</sub> | A <sub>4</sub> |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                              |     |                   |     |                                   |     |                                                 |     |         |     |      |       |    |
| 1      | 1                                               |    | A <sub>3</sub>  | A <sub>2</sub>  | A <sub>1</sub> | A <sub>0</sub> | 0              | 0              | 0              | 0              |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                              |     |                   |     |                                   |     |                                                 |     |         |     |      |       |    |
| 0      | 0                                               | 1C | 0               | 0               | 0              | 1              | 1              | 1              | 0              | 0              | Temperature Sensor Control (Write Command to External temperature sensor) | Write Command to External temperature sensor.<br>A[7:0] = 00h [POR],<br>B[7:0] = 00h [POR],<br>C[7:0] = 00h [POR],<br><br>A[7:6] <table><tr><td>A[7:6]</td><td>Select no of byte to be sent</td></tr><tr><td>00</td><td>Address + pointer</td></tr><tr><td>01</td><td>Address + pointer + 1st parameter</td></tr><tr><td>10</td><td>Address + pointer + 1st parameter + 2nd pointer</td></tr><tr><td>11</td><td>Address</td></tr></table><br>A[5:0] – Pointer Setting<br>B[7:0] – 1 <sup>st</sup> parameter<br>C[7:0] – 2 <sup>nd</sup> parameter<br>The command required CLKEN=1.<br>Refer to Register 0x22 for detail.<br><br>After this command initiated, Write Command to external temperature sensor starts. BUSY pad will output high during operation. | A[7:6] | Select no of byte to be sent | 00  | Address + pointer | 01  | Address + pointer + 1st parameter | 10  | Address + pointer + 1st parameter + 2nd pointer | 11  | Address |     |      |       |    |
| A[7:6] | Select no of byte to be sent                    |    |                 |                 |                |                |                |                |                |                |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                              |     |                   |     |                                   |     |                                                 |     |         |     |      |       |    |
| 00     | Address + pointer                               |    |                 |                 |                |                |                |                |                |                |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                              |     |                   |     |                                   |     |                                                 |     |         |     |      |       |    |
| 01     | Address + pointer + 1st parameter               |    |                 |                 |                |                |                |                |                |                |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                              |     |                   |     |                                   |     |                                                 |     |         |     |      |       |    |
| 10     | Address + pointer + 1st parameter + 2nd pointer |    |                 |                 |                |                |                |                |                |                |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                              |     |                   |     |                                   |     |                                                 |     |         |     |      |       |    |
| 11     | Address                                         |    |                 |                 |                |                |                |                |                |                |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                              |     |                   |     |                                   |     |                                                 |     |         |     |      |       |    |
| 0      | 1                                               |    | A <sub>7</sub>  | A <sub>6</sub>  | A <sub>5</sub> | A <sub>4</sub> | A <sub>3</sub> | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                              |     |                   |     |                                   |     |                                                 |     |         |     |      |       |    |
| 0      | 1                                               |    | B <sub>7</sub>  | B <sub>6</sub>  | B <sub>5</sub> | B <sub>4</sub> | B <sub>3</sub> | B <sub>2</sub> | B <sub>1</sub> | B <sub>0</sub> |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                              |     |                   |     |                                   |     |                                                 |     |         |     |      |       |    |
| 0      | 1                                               |    | C <sub>7</sub>  | C <sub>6</sub>  | C <sub>5</sub> | C <sub>4</sub> | C <sub>3</sub> | C <sub>2</sub> | C <sub>1</sub> | C <sub>0</sub> |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                              |     |                   |     |                                   |     |                                                 |     |         |     |      |       |    |
|        |                                                 |    |                 |                 |                |                |                |                |                |                |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                              |     |                   |     |                                   |     |                                                 |     |         |     |      |       |    |
| 0      | 0                                               | 20 | 0               | 0               | 1              | 0              | 0              | 0              | 0              | 0              | Master Activation                                                         | Activate Display Update Sequence<br><br>The Display Update Sequence Option is located at R22h.<br><br>BUSY pad will output high during operation. User should not interrupt this operation to avoid corruption of panel images.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                              |     |                   |     |                                   |     |                                                 |     |         |     |      |       |    |

| 0                                                                                                                                        | 0                  | 0  | 21             | 0              | 0              | 1              | 0              | 0              | 0              | 0              | 1                                  | Display Update Control 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | RAM content option for Display Update<br>A[7:0] = 00h [POR]<br>B[7:0] = 00h [POR]<br><br>A[7:4] Red RAM option<br>0000 Normal<br>0100 Bypass RAM content as 0<br>1000 Inverse RAM content<br><br>A[3:0] BW RAM option<br>0000 Normal<br>0100 Bypass RAM content as 0<br>1000 Inverse RAM content<br><br>B[7] Source Output Mode<br>0 Available Source from S0 to S175<br>1 Available Source from S8 to S167 |                    |                     |    |                      |    |                                        |    |                                          |    |                                                                                 |    |                                                                                 |    |                                                                                                             |    |                                                                                                             |    |                                                                                                              |    |                                                                                                              |    |                                                                                                                                          |    |                                                                                                                                          |    |
|------------------------------------------------------------------------------------------------------------------------------------------|--------------------|----|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|---------------------|----|----------------------|----|----------------------------------------|----|------------------------------------------|----|---------------------------------------------------------------------------------|----|---------------------------------------------------------------------------------|----|-------------------------------------------------------------------------------------------------------------|----|-------------------------------------------------------------------------------------------------------------|----|--------------------------------------------------------------------------------------------------------------|----|--------------------------------------------------------------------------------------------------------------|----|------------------------------------------------------------------------------------------------------------------------------------------|----|------------------------------------------------------------------------------------------------------------------------------------------|----|
| 0                                                                                                                                        | 1                  |    | A <sub>7</sub> | A <sub>6</sub> | A <sub>5</sub> | A <sub>4</sub> | A <sub>3</sub> | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                             |                    |                     |    |                      |    |                                        |    |                                          |    |                                                                                 |    |                                                                                 |    |                                                                                                             |    |                                                                                                             |    |                                                                                                              |    |                                                                                                              |    |                                                                                                                                          |    |                                                                                                                                          |    |
| 0                                                                                                                                        | 1                  |    | B <sub>7</sub> | 0              | 0              | 0              | 0              | 0              | 0              | 0              |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                             |                    |                     |    |                      |    |                                        |    |                                          |    |                                                                                 |    |                                                                                 |    |                                                                                                             |    |                                                                                                             |    |                                                                                                              |    |                                                                                                              |    |                                                                                                                                          |    |                                                                                                                                          |    |
| 0                                                                                                                                        | 0                  | 22 | 0              | 0              | 1              | 0              | 0              | 0              | 1              | 0              | Display Update Control 2           | Display Update Sequence Option:<br>Enable the stage for Master Activation<br>A[7:0] = FFh (POR)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                             |                    |                     |    |                      |    |                                        |    |                                          |    |                                                                                 |    |                                                                                 |    |                                                                                                             |    |                                                                                                             |    |                                                                                                              |    |                                                                                                              |    |                                                                                                                                          |    |                                                                                                                                          |    |
| 0                                                                                                                                        | 1                  |    | A <sub>7</sub> | A <sub>6</sub> | A <sub>5</sub> | A <sub>4</sub> | A <sub>3</sub> | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> |                                    | <table><thead><tr><th>Operating sequence</th><th>Parameter (in Hex)</th></tr></thead><tbody><tr><td>Enable clock signal</td><td>80</td></tr><tr><td>Disable clock signal</td><td>01</td></tr><tr><td>Enable clock signal<br/>→ Enable Analog</td><td>C0</td></tr><tr><td>Disable Analog<br/>→ Disable clock signal</td><td>03</td></tr><tr><td>Enable clock signal<br/>→ Load LUT with DISPLAY Mode 1<br/>→ Disable clock signal</td><td>91</td></tr><tr><td>Enable clock signal<br/>→ Load LUT with DISPLAY Mode 2<br/>→ Disable clock signal</td><td>99</td></tr><tr><td>Enable clock signal<br/>→ Load temperature value<br/>→ Load LUT with DISPLAY Mode 1<br/>→ Disable clock signal</td><td>B1</td></tr><tr><td>Enable clock signal<br/>→ Load temperature value<br/>→ Load LUT with DISPLAY Mode 2<br/>→ Disable clock signal</td><td>B9</td></tr><tr><td>Enable clock signal<br/>→ Enable Analog<br/>→ Display with DISPLAY Mode 1<br/>→ Disable Analog<br/>→ Disable OSC</td><td>C7</td></tr><tr><td>Enable clock signal<br/>→ Enable Analog<br/>→ Display with DISPLAY Mode 2<br/>→ Disable Analog<br/>→ Disable OSC</td><td>CF</td></tr><tr><td>Enable clock signal<br/>→ Enable Analog<br/>→ Load temperature value<br/>→ DISPLAY with DISPLAY Mode 1<br/>→ Disable Analog<br/>→ Disable OSC</td><td>F7</td></tr><tr><td>Enable clock signal<br/>→ Enable Analog<br/>→ Load temperature value<br/>→ DISPLAY with DISPLAY Mode 2<br/>→ Disable Analog<br/>→ Disable OSC</td><td>FF</td></tr></tbody></table> | Operating sequence                                                                                                                                                                                                                                                                                                                                                                                          | Parameter (in Hex) | Enable clock signal | 80 | Disable clock signal | 01 | Enable clock signal<br>→ Enable Analog | C0 | Disable Analog<br>→ Disable clock signal | 03 | Enable clock signal<br>→ Load LUT with DISPLAY Mode 1<br>→ Disable clock signal | 91 | Enable clock signal<br>→ Load LUT with DISPLAY Mode 2<br>→ Disable clock signal | 99 | Enable clock signal<br>→ Load temperature value<br>→ Load LUT with DISPLAY Mode 1<br>→ Disable clock signal | B1 | Enable clock signal<br>→ Load temperature value<br>→ Load LUT with DISPLAY Mode 2<br>→ Disable clock signal | B9 | Enable clock signal<br>→ Enable Analog<br>→ Display with DISPLAY Mode 1<br>→ Disable Analog<br>→ Disable OSC | C7 | Enable clock signal<br>→ Enable Analog<br>→ Display with DISPLAY Mode 2<br>→ Disable Analog<br>→ Disable OSC | CF | Enable clock signal<br>→ Enable Analog<br>→ Load temperature value<br>→ DISPLAY with DISPLAY Mode 1<br>→ Disable Analog<br>→ Disable OSC | F7 | Enable clock signal<br>→ Enable Analog<br>→ Load temperature value<br>→ DISPLAY with DISPLAY Mode 2<br>→ Disable Analog<br>→ Disable OSC | FF |
| Operating sequence                                                                                                                       | Parameter (in Hex) |    |                |                |                |                |                |                |                |                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                             |                    |                     |    |                      |    |                                        |    |                                          |    |                                                                                 |    |                                                                                 |    |                                                                                                             |    |                                                                                                             |    |                                                                                                              |    |                                                                                                              |    |                                                                                                                                          |    |                                                                                                                                          |    |
| Enable clock signal                                                                                                                      | 80                 |    |                |                |                |                |                |                |                |                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                             |                    |                     |    |                      |    |                                        |    |                                          |    |                                                                                 |    |                                                                                 |    |                                                                                                             |    |                                                                                                             |    |                                                                                                              |    |                                                                                                              |    |                                                                                                                                          |    |                                                                                                                                          |    |
| Disable clock signal                                                                                                                     | 01                 |    |                |                |                |                |                |                |                |                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                             |                    |                     |    |                      |    |                                        |    |                                          |    |                                                                                 |    |                                                                                 |    |                                                                                                             |    |                                                                                                             |    |                                                                                                              |    |                                                                                                              |    |                                                                                                                                          |    |                                                                                                                                          |    |
| Enable clock signal<br>→ Enable Analog                                                                                                   | C0                 |    |                |                |                |                |                |                |                |                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                             |                    |                     |    |                      |    |                                        |    |                                          |    |                                                                                 |    |                                                                                 |    |                                                                                                             |    |                                                                                                             |    |                                                                                                              |    |                                                                                                              |    |                                                                                                                                          |    |                                                                                                                                          |    |
| Disable Analog<br>→ Disable clock signal                                                                                                 | 03                 |    |                |                |                |                |                |                |                |                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                             |                    |                     |    |                      |    |                                        |    |                                          |    |                                                                                 |    |                                                                                 |    |                                                                                                             |    |                                                                                                             |    |                                                                                                              |    |                                                                                                              |    |                                                                                                                                          |    |                                                                                                                                          |    |
| Enable clock signal<br>→ Load LUT with DISPLAY Mode 1<br>→ Disable clock signal                                                          | 91                 |    |                |                |                |                |                |                |                |                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                             |                    |                     |    |                      |    |                                        |    |                                          |    |                                                                                 |    |                                                                                 |    |                                                                                                             |    |                                                                                                             |    |                                                                                                              |    |                                                                                                              |    |                                                                                                                                          |    |                                                                                                                                          |    |
| Enable clock signal<br>→ Load LUT with DISPLAY Mode 2<br>→ Disable clock signal                                                          | 99                 |    |                |                |                |                |                |                |                |                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                             |                    |                     |    |                      |    |                                        |    |                                          |    |                                                                                 |    |                                                                                 |    |                                                                                                             |    |                                                                                                             |    |                                                                                                              |    |                                                                                                              |    |                                                                                                                                          |    |                                                                                                                                          |    |
| Enable clock signal<br>→ Load temperature value<br>→ Load LUT with DISPLAY Mode 1<br>→ Disable clock signal                              | B1                 |    |                |                |                |                |                |                |                |                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                             |                    |                     |    |                      |    |                                        |    |                                          |    |                                                                                 |    |                                                                                 |    |                                                                                                             |    |                                                                                                             |    |                                                                                                              |    |                                                                                                              |    |                                                                                                                                          |    |                                                                                                                                          |    |
| Enable clock signal<br>→ Load temperature value<br>→ Load LUT with DISPLAY Mode 2<br>→ Disable clock signal                              | B9                 |    |                |                |                |                |                |                |                |                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                             |                    |                     |    |                      |    |                                        |    |                                          |    |                                                                                 |    |                                                                                 |    |                                                                                                             |    |                                                                                                             |    |                                                                                                              |    |                                                                                                              |    |                                                                                                                                          |    |                                                                                                                                          |    |
| Enable clock signal<br>→ Enable Analog<br>→ Display with DISPLAY Mode 1<br>→ Disable Analog<br>→ Disable OSC                             | C7                 |    |                |                |                |                |                |                |                |                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                             |                    |                     |    |                      |    |                                        |    |                                          |    |                                                                                 |    |                                                                                 |    |                                                                                                             |    |                                                                                                             |    |                                                                                                              |    |                                                                                                              |    |                                                                                                                                          |    |                                                                                                                                          |    |
| Enable clock signal<br>→ Enable Analog<br>→ Display with DISPLAY Mode 2<br>→ Disable Analog<br>→ Disable OSC                             | CF                 |    |                |                |                |                |                |                |                |                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                             |                    |                     |    |                      |    |                                        |    |                                          |    |                                                                                 |    |                                                                                 |    |                                                                                                             |    |                                                                                                             |    |                                                                                                              |    |                                                                                                              |    |                                                                                                                                          |    |                                                                                                                                          |    |
| Enable clock signal<br>→ Enable Analog<br>→ Load temperature value<br>→ DISPLAY with DISPLAY Mode 1<br>→ Disable Analog<br>→ Disable OSC | F7                 |    |                |                |                |                |                |                |                |                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                             |                    |                     |    |                      |    |                                        |    |                                          |    |                                                                                 |    |                                                                                 |    |                                                                                                             |    |                                                                                                             |    |                                                                                                              |    |                                                                                                              |    |                                                                                                                                          |    |                                                                                                                                          |    |
| Enable clock signal<br>→ Enable Analog<br>→ Load temperature value<br>→ DISPLAY with DISPLAY Mode 2<br>→ Disable Analog<br>→ Disable OSC | FF                 |    |                |                |                |                |                |                |                |                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                             |                    |                     |    |                      |    |                                        |    |                                          |    |                                                                                 |    |                                                                                 |    |                                                                                                             |    |                                                                                                             |    |                                                                                                              |    |                                                                                                              |    |                                                                                                                                          |    |                                                                                                                                          |    |
| 0                                                                                                                                        | 1                  |    | A <sub>7</sub> | A <sub>6</sub> | A <sub>5</sub> | A <sub>4</sub> | A <sub>3</sub> | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                             |                    |                     |    |                      |    |                                        |    |                                          |    |                                                                                 |    |                                                                                 |    |                                                                                                             |    |                                                                                                             |    |                                                                                                              |    |                                                                                                              |    |                                                                                                                                          |    |                                                                                                                                          |    |
| 0                                                                                                                                        | 0                  | 24 | 0              | 0              | 1              | 0              | 0              | 1              | 0              | 0              | Write RAM (Black White) / RAM 0x24 | After this command, data entries will be written into the BW RAM until another command is written. Address pointers will advance accordingly<br><br>For Write pixel:<br>Content of Write RAM(BW) = 1<br>For Black pixel:<br>Content of Write RAM(BW) = 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                             |                    |                     |    |                      |    |                                        |    |                                          |    |                                                                                 |    |                                                                                 |    |                                                                                                             |    |                                                                                                             |    |                                                                                                              |    |                                                                                                              |    |                                                                                                                                          |    |                                                                                                                                          |    |



| Command Table |      |     |    |    |    |    |                |                |                |                |                                 |                                                                                                                                                                                                                                                                                            |
|---------------|------|-----|----|----|----|----|----------------|----------------|----------------|----------------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R/W#          | D/C# | Hex | D7 | D6 | D5 | D4 | D3             | D2             | D1             | D0             | Command                         | Description                                                                                                                                                                                                                                                                                |
| 0             | 0    | 26  | 0  | 0  | 1  | 0  | 0              | 1              | 1              | 0              | Write RAM (RED)<br>/ RAM 0x26   | After this command, data entries will be written into the RED RAM until another command is written. Address pointers will advance accordingly.<br><br>For Red pixel:<br>Content of Write RAM(RED) = 1<br>For non-Red pixel [Black or White]:<br>Content of Write RAM(RED) = 0              |
| 0             | 0    | 27  | 0  | 0  | 1  | 0  | 0              | 1              | 1              | 1              | Read RAM                        | After this command, data read on the MCU bus will fetch data from RAM. According to parameter of Register 41h to select reading RAM0x24/ RAM0x26, until another command is written. Address pointers will advance accordingly.<br><br>The 1 <sup>st</sup> byte of data read is dummy data. |
| 0             | 0    | 28  | 0  | 0  | 1  | 0  | 1              | 0              | 0              | 0              | VCOM Sense                      | Enter VCOM sensing conditions and hold for duration defined in 29h before reading VCOM value.<br>The sensed VCOM voltage is stored in register<br>The command required CLKEN=1 and ANALOGEN=1<br>Refer to Register 0x22 for detail.<br><br>BUSY pad will output high during operation.     |
| 0             | 0    | 29  | 0  | 0  | 1  | 0  | 1              | 0              | 0              | 1              | VCOM Sense Duration             | Stabling time between entering VCOM sensing mode and reading acquired.<br><br>A[3:0] = 9h, duration = 10s.<br>VCOM sense duration = (A[3:0]+1) sec                                                                                                                                         |
| 0             | 1    |     | 0  | 1  | 0  | 0  | A <sub>3</sub> | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> |                                 |                                                                                                                                                                                                                                                                                            |
| 0             | 0    | 2A  | 0  | 0  | 1  | 0  | 1              | 0              | 1              | 0              | Program VCOM OTP                | Program VCOM register into OTP<br><br>The command required CLKEN=1.<br>Refer to Register 0x22 for detail.<br><br>BUSY pad will output high during operation.                                                                                                                               |
| 0             | 0    | 2B  | 0  | 0  | 1  | 0  | 1              | 0              | 1              | 1              | Write Register for VCOM Control | This command is used to reduce glitch when ACVCOM toggle. Two data bytes D04h and D63h should be set for this command.                                                                                                                                                                     |
| 0             | 1    |     | 0  | 0  | 0  | 0  | 0              | 1              | 0              | 0              |                                 |                                                                                                                                                                                                                                                                                            |
| 0             | 1    |     | 0  | 1  | 1  | 0  | 0              | 0              | 1              | 1              |                                 |                                                                                                                                                                                                                                                                                            |

| Command Table |      |        |                |                |                |                |                |                |                |                |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
|---------------|------|--------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------|--------|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|----|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|----|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|----|-----|------|-------|----|
| R/W#          | D/C# | Hex    | D7             | D6             | D5             | D4             | D3             | D2             | D1             | D0             | Command                              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 0             | 0    | 2C     | 0              | 0              | 1              | 0              | 1              | 1              | 0              | 0              | Write VCOM register                  | Write VCOM register from MCU interface<br>A[7:0] = 00h [POR]<br><table><tr><th>A[7:0]</th><th>VCOM</th><th>A[7:0]</th><th>VCOM</th></tr><tr><td>08h</td><td>-0.2</td><td>44h</td><td>-1.7</td></tr><tr><td>0Ch</td><td>-0.3</td><td>48h</td><td>-1.8</td></tr><tr><td>10h</td><td>-0.4</td><td>4Ch</td><td>-1.9</td></tr><tr><td>14h</td><td>-0.5</td><td>50h</td><td>-2</td></tr><tr><td>18h</td><td>-0.6</td><td>54h</td><td>-2.1</td></tr><tr><td>1Ch</td><td>-0.7</td><td>58h</td><td>-2.2</td></tr><tr><td>20h</td><td>-0.8</td><td>5Ch</td><td>-2.3</td></tr><tr><td>24h</td><td>-0.9</td><td>60h</td><td>-2.4</td></tr><tr><td>28h</td><td>-1</td><td>64h</td><td>-2.5</td></tr><tr><td>2Ch</td><td>-1.1</td><td>68h</td><td>-2.6</td></tr><tr><td>30h</td><td>-1.2</td><td>6Ch</td><td>-2.7</td></tr><tr><td>34h</td><td>-1.3</td><td>70h</td><td>-2.8</td></tr><tr><td>38h</td><td>-1.4</td><td>74h</td><td>-2.9</td></tr><tr><td>3Ch</td><td>-1.5</td><td>78h</td><td>-3</td></tr><tr><td>40h</td><td>-1.6</td><td>Other</td><td>NA</td></tr></table> | A[7:0] | VCOM | A[7:0] | VCOM | 08h | -0.2 | 44h | -1.7 | 0Ch | -0.3 | 48h | -1.8 | 10h | -0.4 | 4Ch | -1.9 | 14h | -0.5 | 50h | -2 | 18h | -0.6 | 54h | -2.1 | 1Ch | -0.7 | 58h | -2.2 | 20h | -0.8 | 5Ch | -2.3 | 24h | -0.9 | 60h | -2.4 | 28h | -1 | 64h | -2.5 | 2Ch | -1.1 | 68h | -2.6 | 30h | -1.2 | 6Ch | -2.7 | 34h | -1.3 | 70h | -2.8 | 38h | -1.4 | 74h | -2.9 | 3Ch | -1.5 | 78h | -3 | 40h | -1.6 | Other | NA |
| A[7:0]        | VCOM | A[7:0] | VCOM           |                |                |                |                |                |                |                |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 08h           | -0.2 | 44h    | -1.7           |                |                |                |                |                |                |                |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 0Ch           | -0.3 | 48h    | -1.8           |                |                |                |                |                |                |                |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 10h           | -0.4 | 4Ch    | -1.9           |                |                |                |                |                |                |                |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 14h           | -0.5 | 50h    | -2             |                |                |                |                |                |                |                |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 18h           | -0.6 | 54h    | -2.1           |                |                |                |                |                |                |                |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 1Ch           | -0.7 | 58h    | -2.2           |                |                |                |                |                |                |                |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 20h           | -0.8 | 5Ch    | -2.3           |                |                |                |                |                |                |                |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 24h           | -0.9 | 60h    | -2.4           |                |                |                |                |                |                |                |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 28h           | -1   | 64h    | -2.5           |                |                |                |                |                |                |                |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 2Ch           | -1.1 | 68h    | -2.6           |                |                |                |                |                |                |                |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 30h           | -1.2 | 6Ch    | -2.7           |                |                |                |                |                |                |                |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 34h           | -1.3 | 70h    | -2.8           |                |                |                |                |                |                |                |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 38h           | -1.4 | 74h    | -2.9           |                |                |                |                |                |                |                |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 3Ch           | -1.5 | 78h    | -3             |                |                |                |                |                |                |                |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 40h           | -1.6 | Other  | NA             |                |                |                |                |                |                |                |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 0             | 1    |        | A <sub>7</sub> | A <sub>6</sub> | A <sub>5</sub> | A <sub>4</sub> | A <sub>3</sub> | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
|               |      |        |                |                |                |                |                |                |                |                |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 0             | 0    | 2D     | 0              | 0              | 1              | 0              | 1              | 1              | 0              | 1              | OTP Register Read for Display Option | Read Register for Display Option:<br><br>A[7:0]: VCOM OTP Selection<br>(Command 0x37, Byte A)<br><br>B[7:0]: VCOM Register<br>(Command 0x2C)<br><br>C[7:0]~G[7:0]: Display Mode<br>(Command 0x37, Byte B to Byte F)<br>[5 bytes]<br><br>H[7:0]~K[7:0]: Waveform Version<br>(Command 0x37, Byte G to Byte J)<br>[4 bytes]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 1             | 1    |        | A <sub>7</sub> | A <sub>6</sub> | A <sub>5</sub> | A <sub>4</sub> | A <sub>3</sub> | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 1             | 1    |        | B <sub>7</sub> | B <sub>6</sub> | B <sub>5</sub> | B <sub>4</sub> | B <sub>3</sub> | B <sub>2</sub> | B <sub>1</sub> | B <sub>0</sub> |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 1             | 1    |        | C <sub>7</sub> | C <sub>6</sub> | C <sub>5</sub> | C <sub>4</sub> | C <sub>3</sub> | C <sub>2</sub> | C <sub>1</sub> | C <sub>0</sub> |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 1             | 1    |        | D <sub>7</sub> | D <sub>6</sub> | D <sub>5</sub> | D <sub>4</sub> | D <sub>3</sub> | D <sub>2</sub> | D <sub>1</sub> | D <sub>0</sub> |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 1             | 1    |        | E <sub>7</sub> | E <sub>6</sub> | E <sub>5</sub> | E <sub>4</sub> | E <sub>3</sub> | E <sub>2</sub> | E <sub>1</sub> | E <sub>0</sub> |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 1             | 1    |        | F <sub>7</sub> | F <sub>6</sub> | F <sub>5</sub> | F <sub>4</sub> | F <sub>3</sub> | F <sub>2</sub> | F <sub>1</sub> | F <sub>0</sub> |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 1             | 1    |        | G <sub>7</sub> | G <sub>6</sub> | G <sub>5</sub> | G <sub>4</sub> | G <sub>3</sub> | G <sub>2</sub> | G <sub>1</sub> | G <sub>0</sub> |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 1             | 1    |        | H <sub>7</sub> | H <sub>6</sub> | H <sub>5</sub> | H <sub>4</sub> | H <sub>3</sub> | H <sub>2</sub> | H <sub>1</sub> | H <sub>0</sub> |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 1             | 1    |        | I <sub>7</sub> | I <sub>6</sub> | I <sub>5</sub> | I <sub>4</sub> | I <sub>3</sub> | I <sub>2</sub> | I <sub>1</sub> | I <sub>0</sub> |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 1             | 1    |        | J <sub>7</sub> | J <sub>6</sub> | J <sub>5</sub> | J <sub>4</sub> | J <sub>3</sub> | J <sub>2</sub> | J <sub>1</sub> | J <sub>0</sub> |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 1             | 1    |        | K <sub>7</sub> | K <sub>6</sub> | K <sub>5</sub> | K <sub>4</sub> | K <sub>3</sub> | K <sub>2</sub> | K <sub>1</sub> | K <sub>0</sub> |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 0             | 0    | 2E     | 0              | 0              | 1              | 0              | 1              | 1              | 1              | 0              | User ID Read                         | Read 10 Byte User ID stored in OTP:<br>A[7:0]~J[7:0]: UserID (R38, Byte A and Byte J) [10 bytes]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 1             | 1    |        | A <sub>7</sub> | A <sub>6</sub> | A <sub>5</sub> | A <sub>4</sub> | A <sub>3</sub> | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 1             | 1    |        | B <sub>7</sub> | B <sub>6</sub> | B <sub>5</sub> | B <sub>4</sub> | B <sub>3</sub> | B <sub>2</sub> | B <sub>1</sub> | B <sub>0</sub> |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 1             | 1    |        | C <sub>7</sub> | C <sub>6</sub> | C <sub>5</sub> | C <sub>4</sub> | C <sub>3</sub> | C <sub>2</sub> | C <sub>1</sub> | C <sub>0</sub> |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 1             | 1    |        | D <sub>7</sub> | D <sub>6</sub> | D <sub>5</sub> | D <sub>4</sub> | D <sub>3</sub> | D <sub>2</sub> | D <sub>1</sub> | D <sub>0</sub> |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 1             | 1    |        | E <sub>7</sub> | E <sub>6</sub> | E <sub>5</sub> | E <sub>4</sub> | E <sub>3</sub> | E <sub>2</sub> | E <sub>1</sub> | E <sub>0</sub> |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 1             | 1    |        | F <sub>7</sub> | F <sub>6</sub> | F <sub>5</sub> | F <sub>4</sub> | F <sub>3</sub> | F <sub>2</sub> | F <sub>1</sub> | F <sub>0</sub> |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 1             | 1    |        | G <sub>7</sub> | G <sub>6</sub> | G <sub>5</sub> | G <sub>4</sub> | G <sub>3</sub> | G <sub>2</sub> | G <sub>1</sub> | G <sub>0</sub> |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 1             | 1    |        | H <sub>7</sub> | H <sub>6</sub> | H <sub>5</sub> | H <sub>4</sub> | H <sub>3</sub> | H <sub>2</sub> | H <sub>1</sub> | H <sub>0</sub> |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 1             | 1    |        | I <sub>7</sub> | I <sub>6</sub> | I <sub>5</sub> | I <sub>4</sub> | I <sub>3</sub> | I <sub>2</sub> | I <sub>1</sub> | I <sub>0</sub> |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 1             | 1    |        | J <sub>7</sub> | J <sub>6</sub> | J <sub>5</sub> | J <sub>4</sub> | J <sub>3</sub> | J <sub>2</sub> | J <sub>1</sub> | J <sub>0</sub> |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 0             | 0    | 2F     | 0              | 0              | 1              | 0              | 1              | 1              | 1              | 1              | Status Bit Read                      | Read IC status Bit [POR 0x01]<br>A[5]: HV Ready Detection flag [POR=0]<br>0: Ready<br>1: Not Ready<br>A[4]: VCI Detection flag [POR=0]<br>0: Normal<br>1: VCI lower than the Detect level<br>A[3]: [POR=0]<br>A[2]: Busy flag [POR=0]<br>0: Normal<br>1: BUSY<br>A[1:0]: Chip ID [POR=01]<br><br>Remark:<br>A[5] and A[4] status are not valid after RESET, they need to be initiated by command 0x14 and command 0x15 respectively.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |
| 1             | 1    |        | 0              | 0              | A <sub>5</sub> | A <sub>4</sub> | 0              | 0              | A <sub>1</sub> | A <sub>0</sub> |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |        |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |    |     |      |       |    |

|   |   |    |                 |                 |                 |                 |                 |                 |                |                |                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---|---|----|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|----------------|----------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0 | 0 | 30 | 0               | 0               | 1               | 1               | 0               | 0               | 0              | 0              | Program WS OTP                    | Program OTP of Waveform Setting<br>The contents should be written into RAM before sending this command.<br><br>The command required CLKEN=1. Refer to Register 0x22 for detail. BUSY pad will output high during operation.                                                                                                                                                                                                                                                                                                                                                                                            |
| 0 | 0 | 31 | 0               | 0               | 1               | 1               | 0               | 0               | 0              | 1              | Load WS OTP                       | Load OTP of Waveform Setting<br><br>The command required CLKEN=1. Refer to Register 0x22 for detail.<br><br>BUSY pad will output high during operation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 0 | 0 | 32 | 0               | 0               | 1               | 1               | 0               | 0               | 1              | 0              | Write LUT register                | Write LUT register from MCU interface [153 bytes], which contains the content of VS[nX-LUTm], TP[nX], RP[n], SR[nXY], FR[n] and XON[nXY]<br>Refer to Session 6.7 WAVEFORM SETTING                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 0 | 1 |    | A <sub>7</sub>  | A <sub>6</sub>  | A <sub>5</sub>  | A <sub>4</sub>  | A <sub>3</sub>  | A <sub>2</sub>  | A <sub>1</sub> | A <sub>0</sub> |                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 0 | 1 |    | B <sub>7</sub>  | B <sub>6</sub>  | B <sub>5</sub>  | B <sub>4</sub>  | B <sub>3</sub>  | B <sub>2</sub>  | B <sub>1</sub> | B <sub>0</sub> |                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 0 | 1 |    | :               | :               | :               | :               | :               | :               | :              | :              |                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 0 | 1 |    | .               | ..              | .               | .               | .               | .               | .              | .              |                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 0 | 0 | 34 | 0               | 0               | 1               | 1               | 0               | 1               | 0              | 0              | CRC calculation                   | CRC calculation command<br>For details, please refer to SSD1680 application note.<br><br>BUSY pad will output high during operation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 0 | 0 | 35 | 0               | 0               | 1               | 1               | 0               | 1               | 0              | 1              | CRC Status Read                   | CRC Status Read<br>A[15:0] is the CRC read out value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 1 | 1 |    | A <sub>15</sub> | A <sub>14</sub> | A <sub>13</sub> | A <sub>12</sub> | A <sub>11</sub> | A <sub>10</sub> | A <sub>9</sub> | A <sub>8</sub> |                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 1 | 1 |    | A <sub>7</sub>  | A <sub>6</sub>  | A <sub>5</sub>  | A <sub>4</sub>  | A <sub>3</sub>  | A <sub>2</sub>  | A <sub>1</sub> | A <sub>0</sub> |                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 0 | 0 | 36 | 0               | 0               | 1               | 1               | 0               | 1               | 1              | 0              | Program OTP selection             | Program OTP Selection according to the OTP Selection Control [R37h and R38h]<br><br>The command required CLKEN=1. Refer to Register 0x22 for detail. BUSY pad will output high during operation.                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 0 | 0 | 37 | 0               | 0               | 1               | 1               | 0               | 1               | 1              | 1              | Write Register for Display Option | Write Register for Display Option<br>A[7] Spare VCOM OTP selection<br>0: Default [POR]<br>1: Spare<br><br>B[7:0] Display Mode for WS[7:0]<br>C[7:0] Display Mode for WS[15:8]<br>D[7:0] Display Mode for WS[23:16]<br>E[7:0] Display Mode for WS[31:24]<br>F[3:0] Display Mode for WS[35:32]<br>0: Display Mode 1<br>1: Display Mode 2<br><br>F[6]: PingPong for Display Mode 2<br>0: RAM Ping-Pong disable [POR]<br>1: RAM Ping-Pong enable<br><br>G[7:0]~J[7:0] module ID /waveform version.<br><br>Remarks:<br>1) A[7:0]~J[7:0] can be stored in OTP<br>2) RAM Ping-Pong function is not support for Display Mode 1 |
| 0 | 1 |    | A <sub>7</sub>  | 0               | 0               | 0               | 0               | 0               | 0              | 0              |                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 0 | 1 |    | B <sub>7</sub>  | B <sub>6</sub>  | B <sub>5</sub>  | B <sub>4</sub>  | B <sub>3</sub>  | B <sub>2</sub>  | B <sub>1</sub> | B <sub>0</sub> |                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 0 | 1 |    | C <sub>7</sub>  | C <sub>6</sub>  | C <sub>5</sub>  | C <sub>4</sub>  | C <sub>3</sub>  | C <sub>2</sub>  | C <sub>1</sub> | C <sub>0</sub> |                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 0 | 1 |    | D <sub>7</sub>  | D <sub>6</sub>  | D <sub>5</sub>  | D <sub>4</sub>  | D <sub>3</sub>  | D <sub>2</sub>  | D <sub>1</sub> | D <sub>0</sub> |                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 0 | 1 |    | E <sub>7</sub>  | E <sub>6</sub>  | E <sub>5</sub>  | E <sub>4</sub>  | E <sub>3</sub>  | E <sub>2</sub>  | E <sub>1</sub> | E <sub>0</sub> |                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 0 | 1 |    | 0               | F <sub>6</sub>  | 0               | 0               | F <sub>3</sub>  | F <sub>2</sub>  | F <sub>1</sub> | F <sub>0</sub> |                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 0 | 1 |    | G <sub>7</sub>  | G <sub>6</sub>  | G <sub>5</sub>  | G <sub>4</sub>  | G <sub>3</sub>  | G <sub>2</sub>  | G <sub>1</sub> | G <sub>0</sub> |                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 0 | 1 |    | H <sub>7</sub>  | H <sub>6</sub>  | H <sub>5</sub>  | H <sub>4</sub>  | H <sub>3</sub>  | H <sub>2</sub>  | H <sub>1</sub> | H <sub>0</sub> |                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 0 | 1 |    | I <sub>7</sub>  | I <sub>6</sub>  | I <sub>5</sub>  | I <sub>4</sub>  | I <sub>3</sub>  | I <sub>2</sub>  | I <sub>1</sub> | I <sub>0</sub> |                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 0 | 1 |    | J <sub>7</sub>  | J <sub>6</sub>  | J <sub>5</sub>  | J <sub>4</sub>  | J <sub>3</sub>  | J <sub>2</sub>  | J <sub>1</sub> | J <sub>0</sub> |                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |



|         |                                                           |    |                |                |                |                |                |                |                |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
|---------|-----------------------------------------------------------|----|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------------|-----|-----------------------------------------------------------|----|------------------------------|----|------|---------|-----|--------|-----------|----|-----|----|------|----|-----|----|------|------|-----------------------|---|--------------------------------|---|------------|--------|----------------|----|------|----|------|----|------|----|------|
| 0       | 0                                                         | 38 | 0              | 0              | 1              | 1              | 1              | 0              | 0              | 0              | Write Register for User ID<br>A[7:0]~J[7:0]: UserID [10 bytes]<br><br>Remarks: A[7:0]~J[7:0] can be stored in OTP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 0       | 1                                                         |    | A <sub>7</sub> | A <sub>6</sub> | A <sub>5</sub> | A <sub>4</sub> | A <sub>3</sub> | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 0       | 1                                                         |    | B <sub>7</sub> | B <sub>6</sub> | B <sub>5</sub> | B <sub>4</sub> | B <sub>3</sub> | B <sub>2</sub> | B <sub>1</sub> | B <sub>0</sub> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 0       | 1                                                         |    | C <sub>7</sub> | C <sub>6</sub> | C <sub>5</sub> | C <sub>4</sub> | C <sub>3</sub> | C <sub>2</sub> | C <sub>1</sub> | C <sub>0</sub> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 0       | 1                                                         |    | D <sub>7</sub> | D <sub>6</sub> | D <sub>5</sub> | D <sub>4</sub> | D <sub>3</sub> | D <sub>2</sub> | D <sub>1</sub> | D <sub>0</sub> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 0       | 1                                                         |    | E <sub>7</sub> | E <sub>6</sub> | E <sub>5</sub> | E <sub>4</sub> | E <sub>3</sub> | E <sub>2</sub> | E <sub>1</sub> | E <sub>0</sub> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 0       | 1                                                         |    | F <sub>7</sub> | F <sub>6</sub> | F <sub>5</sub> | F <sub>4</sub> | F <sub>3</sub> | F <sub>2</sub> | F <sub>1</sub> | F <sub>0</sub> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 0       | 1                                                         |    | G <sub>7</sub> | G <sub>6</sub> | G <sub>5</sub> | G <sub>4</sub> | G <sub>3</sub> | G <sub>2</sub> | G <sub>1</sub> | G <sub>0</sub> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 0       | 1                                                         |    | H <sub>7</sub> | H <sub>6</sub> | H <sub>5</sub> | H <sub>4</sub> | H <sub>3</sub> | H <sub>2</sub> | H <sub>1</sub> | H <sub>0</sub> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 0       | 1                                                         |    | I <sub>7</sub> | I <sub>6</sub> | I <sub>5</sub> | I <sub>4</sub> | I <sub>3</sub> | I <sub>2</sub> | I <sub>1</sub> | I <sub>0</sub> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 0       | 1                                                         |    | J <sub>7</sub> | J <sub>6</sub> | J <sub>5</sub> | J <sub>4</sub> | J <sub>3</sub> | J <sub>2</sub> | J <sub>1</sub> | J <sub>0</sub> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 0       | 0                                                         | 39 | 0              | 0              | 1              | 1              | 1              | 0              | 0              | 1              | OTP program mode<br>A[1:0] = 00: Normal Mode [POR]<br>A[1:0] = 11: Internal generated OTP programming voltage<br><br>Remark: User is required to EXACTLY follow the reference code sequences                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 0       | 1                                                         |    | 0              | 0              | 0              | 0              | 0              | 0              | A <sub>1</sub> | A <sub>0</sub> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 0       | 0                                                         | 3C | 0              | 0              | 1              | 1              | 1              | 1              | 0              | 0              | Border Waveform Control<br>Select border waveform for VBD<br>A[7:0] = C0h [POR], set VBD as HiZ.<br>A [7:6] :Select VBD option<br><table><tr><td>A[7:6]</td><td>Select VBD as</td></tr><tr><td>00</td><td>GS Transition, Defined in A[2] and A[1:0]</td></tr><tr><td>01</td><td>Fix Level, Defined in A[5:4]</td></tr><tr><td>10</td><td>VCOM</td></tr><tr><td>11[POR]</td><td>HiZ</td></tr></table><br>A [5:4] Fix Level Setting for VBD<br><table><tr><td>A[5:4]</td><td>VBD level</td></tr><tr><td>00</td><td>VSS</td></tr><tr><td>01</td><td>VSH1</td></tr><tr><td>10</td><td>VSL</td></tr><tr><td>11</td><td>VSH2</td></tr></table><br>A[2] GS Transition control<br><table><tr><td>A[2]</td><td>GS Transition control</td></tr><tr><td>0</td><td>Follow LUT (Output VCOM @ RED)</td></tr><tr><td>1</td><td>Follow LUT</td></tr></table><br>A [1:0] GS Transition setting for VBD<br><table><tr><td>A[1:0]</td><td>VBD Transition</td></tr><tr><td>00</td><td>LUT0</td></tr><tr><td>01</td><td>LUT1</td></tr><tr><td>10</td><td>LUT2</td></tr><tr><td>11</td><td>LUT3</td></tr></table> | A[7:6] | Select VBD as | 00  | GS Transition, Defined in A[2] and A[1:0]                 | 01 | Fix Level, Defined in A[5:4] | 10 | VCOM | 11[POR] | HiZ | A[5:4] | VBD level | 00 | VSS | 01 | VSH1 | 10 | VSL | 11 | VSH2 | A[2] | GS Transition control | 0 | Follow LUT (Output VCOM @ RED) | 1 | Follow LUT | A[1:0] | VBD Transition | 00 | LUT0 | 01 | LUT1 | 10 | LUT2 | 11 | LUT3 |
| A[7:6]  | Select VBD as                                             |    |                |                |                |                |                |                |                |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 00      | GS Transition, Defined in A[2] and A[1:0]                 |    |                |                |                |                |                |                |                |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 01      | Fix Level, Defined in A[5:4]                              |    |                |                |                |                |                |                |                |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 10      | VCOM                                                      |    |                |                |                |                |                |                |                |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 11[POR] | HiZ                                                       |    |                |                |                |                |                |                |                |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| A[5:4]  | VBD level                                                 |    |                |                |                |                |                |                |                |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 00      | VSS                                                       |    |                |                |                |                |                |                |                |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 01      | VSH1                                                      |    |                |                |                |                |                |                |                |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 10      | VSL                                                       |    |                |                |                |                |                |                |                |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 11      | VSH2                                                      |    |                |                |                |                |                |                |                |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| A[2]    | GS Transition control                                     |    |                |                |                |                |                |                |                |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 0       | Follow LUT (Output VCOM @ RED)                            |    |                |                |                |                |                |                |                |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 1       | Follow LUT                                                |    |                |                |                |                |                |                |                |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| A[1:0]  | VBD Transition                                            |    |                |                |                |                |                |                |                |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 00      | LUT0                                                      |    |                |                |                |                |                |                |                |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 01      | LUT1                                                      |    |                |                |                |                |                |                |                |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 10      | LUT2                                                      |    |                |                |                |                |                |                |                |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 11      | LUT3                                                      |    |                |                |                |                |                |                |                |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 0       | 1                                                         |    | A <sub>7</sub> | A <sub>6</sub> | A <sub>5</sub> | A <sub>4</sub> | 0              | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 0       | 0                                                         | 3F | 0              | 0              | 1              | 1              | 1              | 1              | 1              | 1              | End Option (EOPT)<br>Option for LUT end<br>A[7:0]= 02h [POR]<br><table><tr><td>22h</td><td>Normal.</td></tr><tr><td>07h</td><td>Source output level keep previous output before power off</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 22h    | Normal.       | 07h | Source output level keep previous output before power off |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 22h     | Normal.                                                   |    |                |                |                |                |                |                |                |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 07h     | Source output level keep previous output before power off |    |                |                |                |                |                |                |                |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 0       | 1                                                         |    | A <sub>7</sub> | A <sub>6</sub> | A <sub>5</sub> | A <sub>4</sub> | A <sub>3</sub> | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 0       | 0                                                         | 41 | 0              | 1              | 0              | 0              | 0              | 0              | 0              | 1              | Read RAM Option<br>A[0]= 0 [POR]<br>0 : Read RAM corresponding to RAM0x24<br>1 : Read RAM corresponding to RAM0x26                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 0       | 1                                                         |    | 0              | 0              | 0              | 0              | 0              | 0              | 0              | A <sub>0</sub> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 0       | 0                                                         | 44 | 0              | 1              | 0              | 0              | 0              | 1              | 0              | 0              | Set RAM X - address<br>Start / End position<br><br>Specify the start/end positions of the window address in the X direction by an address unit for RAM<br><br>A[5:0]: XSA[5:0], XStart, POR = 00h<br>B[5:0]: XEA[5:0], XEnd, POR = 15h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 0       | 1                                                         |    | 0              | 0              | A <sub>5</sub> | A <sub>4</sub> | A <sub>3</sub> | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |
| 0       | 1                                                         |    | 0              | 0              | B <sub>5</sub> | B <sub>4</sub> | B <sub>3</sub> | B <sub>2</sub> | B <sub>1</sub> | B <sub>0</sub> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |               |     |                                                           |    |                              |    |      |         |     |        |           |    |     |    |      |    |     |    |      |      |                       |   |                                |   |            |        |                |    |      |    |      |    |      |    |      |

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|--------|--------|--------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|--------|--------|-----|---|-----|-----|-----|----|-----|-----|-----|----|-----|-----|-----|----|-----|----|--------|-------|--------|-------|-----|---|-----|-----|-----|----|-----|-----|-----|----|-----|----|-----|----|-----|----|
| 0      | 0      | 45     | 0              | 1              | 0              | 0              | 0              | 1              | 0              | 1              | Set Ram Y- address<br>Start / End position | Specify the start/end positions of the window address in the Y direction by an address unit for RAM<br><br>A[8:0]: YSA[8:0], YStart, POR = 000h<br>B[8:0]: YEA[8:0], YEnd, POR = 127h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 0      | 1      |        | A <sub>7</sub> | A <sub>6</sub> | A <sub>5</sub> | A <sub>4</sub> | A <sub>3</sub> | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 0      | 1      |        | 0              | 0              | 0              | 0              | 0              | 0              | 0              | A <sub>8</sub> |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 0      | 1      |        | B <sub>7</sub> | B <sub>6</sub> | B <sub>5</sub> | B <sub>4</sub> | B <sub>3</sub> | B <sub>2</sub> | B <sub>1</sub> | B <sub>0</sub> |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 0      | 1      |        | 0              | 0              | 0              | 0              | 0              | 0              | 0              | B <sub>8</sub> |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 0      | 0      | 46     | 0              | 1              | 0              | 0              | 0              | 1              | 1              | 0              | Auto Write RED RAM for<br>Regular Pattern  | Auto Write RED RAM for Regular Pattern<br>A[7:0] = 00h [POR]<br><br>A[7]: The 1st step value, POR = 0<br>A[6:4]: Step Height, POR= 000<br>Step of alter RAM in Y-direction according to Gate<br><table><tr><td>A[6:4]</td><td>Height</td><td>A[6:4]</td><td>Height</td></tr><tr><td>000</td><td>8</td><td>100</td><td>128</td></tr><tr><td>001</td><td>16</td><td>101</td><td>256</td></tr><tr><td>010</td><td>32</td><td>110</td><td>296</td></tr><tr><td>011</td><td>64</td><td>111</td><td>NA</td></tr></table><br>A[2:0]: Step Width, POR= 000<br>Step of alter RAM in X-direction according to Source<br><table><tr><td>A[2:0]</td><td>Width</td><td>A[2:0]</td><td>Width</td></tr><tr><td>000</td><td>8</td><td>100</td><td>128</td></tr><tr><td>001</td><td>16</td><td>101</td><td>176</td></tr><tr><td>010</td><td>32</td><td>110</td><td>NA</td></tr><tr><td>011</td><td>64</td><td>111</td><td>NA</td></tr></table><br>BUSY pad will output high during operation.  | A[6:4] | Height | A[6:4] | Height | 000 | 8 | 100 | 128 | 001 | 16 | 101 | 256 | 010 | 32 | 110 | 296 | 011 | 64 | 111 | NA | A[2:0] | Width | A[2:0] | Width | 000 | 8 | 100 | 128 | 001 | 16 | 101 | 176 | 010 | 32 | 110 | NA | 011 | 64 | 111 | NA |
| A[6:4] | Height | A[6:4] | Height         |                |                |                |                |                |                |                |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 000    | 8      | 100    | 128            |                |                |                |                |                |                |                |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 001    | 16     | 101    | 256            |                |                |                |                |                |                |                |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 010    | 32     | 110    | 296            |                |                |                |                |                |                |                |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 011    | 64     | 111    | NA             |                |                |                |                |                |                |                |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| A[2:0] | Width  | A[2:0] | Width          |                |                |                |                |                |                |                |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 000    | 8      | 100    | 128            |                |                |                |                |                |                |                |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 001    | 16     | 101    | 176            |                |                |                |                |                |                |                |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 010    | 32     | 110    | NA             |                |                |                |                |                |                |                |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 011    | 64     | 111    | NA             |                |                |                |                |                |                |                |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 0      | 1      |        | A <sub>7</sub> | A <sub>6</sub> | A <sub>5</sub> | A <sub>4</sub> | 0              | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
|        |        |        |                |                |                |                |                |                |                |                |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
|        |        |        |                |                |                |                |                |                |                |                |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 0      | 0      | 47     | 0              | 1              | 0              | 0              | 0              | 1              | 1              | 1              | Auto Write B/W RAM for<br>Regular Pattern  | Auto Write B/W RAM for Regular Pattern<br>A[7:0] = 00h [POR]<br><br>A[7]: The 1st step value, POR = 0<br>A[6:4]: Step Height, POR= 000<br>Step of alter RAM in Y-direction according to Gate<br><table><tr><td>A[6:4]</td><td>Height</td><td>A[6:4]</td><td>Height</td></tr><tr><td>000</td><td>8</td><td>100</td><td>128</td></tr><tr><td>001</td><td>16</td><td>101</td><td>256</td></tr><tr><td>010</td><td>32</td><td>110</td><td>296</td></tr><tr><td>011</td><td>64</td><td>111</td><td>NA</td></tr></table><br>A[2:0]: Step Width, POR= 000<br>Step of alter RAM in X-direction according to Source<br><table><tr><td>A[2:0]</td><td>Width</td><td>A[2:0]</td><td>Width</td></tr><tr><td>000</td><td>8</td><td>100</td><td>128</td></tr><tr><td>001</td><td>16</td><td>101</td><td>176</td></tr><tr><td>010</td><td>32</td><td>110</td><td>NA</td></tr><tr><td>011</td><td>64</td><td>111</td><td>NA</td></tr></table><br>During operation, BUSY pad will output high. | A[6:4] | Height | A[6:4] | Height | 000 | 8 | 100 | 128 | 001 | 16 | 101 | 256 | 010 | 32 | 110 | 296 | 011 | 64 | 111 | NA | A[2:0] | Width | A[2:0] | Width | 000 | 8 | 100 | 128 | 001 | 16 | 101 | 176 | 010 | 32 | 110 | NA | 011 | 64 | 111 | NA |
| A[6:4] | Height | A[6:4] | Height         |                |                |                |                |                |                |                |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 000    | 8      | 100    | 128            |                |                |                |                |                |                |                |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 001    | 16     | 101    | 256            |                |                |                |                |                |                |                |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 010    | 32     | 110    | 296            |                |                |                |                |                |                |                |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 011    | 64     | 111    | NA             |                |                |                |                |                |                |                |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| A[2:0] | Width  | A[2:0] | Width          |                |                |                |                |                |                |                |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 000    | 8      | 100    | 128            |                |                |                |                |                |                |                |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 001    | 16     | 101    | 176            |                |                |                |                |                |                |                |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 010    | 32     | 110    | NA             |                |                |                |                |                |                |                |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 011    | 64     | 111    | NA             |                |                |                |                |                |                |                |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 0      | 1      |        | A <sub>7</sub> | A <sub>6</sub> | A <sub>5</sub> | A <sub>4</sub> | 0              | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
|        |        |        |                |                |                |                |                |                |                |                |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
|        |        |        |                |                |                |                |                |                |                |                |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 0      | 0      | 4E     | 0              | 1              | 0              | 0              | 1              | 1              | 1              | 0              | Set RAM X address<br>counter               | Make initial settings for the RAM X address in the address counter (AC)<br>A[5:0]: 00h [POR].                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 0      | 1      |        | 0              | 0              | A <sub>5</sub> | A <sub>4</sub> | A <sub>3</sub> | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 0      | 0      | 4F     | 0              | 1              | 0              | 0              | 1              | 1              | 1              | 1              | Set RAM Y address<br>counter               | Make initial settings for the RAM Y address in the address counter (AC)<br>A[8:0]: 000h [POR].                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 0      | 1      |        | A <sub>7</sub> | A <sub>6</sub> | A <sub>5</sub> | A <sub>4</sub> | A <sub>3</sub> | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 0      | 1      |        | 0              | 0              | 0              | 0              | 0              | 0              | 0              | A <sub>8</sub> |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
| 0      | 0      | 7F     | 0              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | NOP                                        | This command is an empty command; it does not have any effect on the display module.<br>However it can be used to terminate Frame Memory Write or Read Commands.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |
|        |        |        |                |                |                |                |                |                |                |                |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |        |        |        |     |   |     |     |     |    |     |     |     |    |     |     |     |    |     |    |        |       |        |       |     |   |     |     |     |    |     |     |     |    |     |    |     |    |     |    |



## 8.Optical Specifications

Measurements are made with that the illumination is under an angle of 45 degree, the detection is perpendicular unless otherwise specified

| Symbol   | Parameter          | Conditions | Min | Typ.                   | Max | Units | Notes |
|----------|--------------------|------------|-----|------------------------|-----|-------|-------|
| R        | White Reflectivity | White      | 30  | 35                     | -   | %     | 8-1   |
| CR       | Contrast Ratio     | Indoor     | 8:1 |                        | -   |       | 8-2   |
| GN       | 2Grey Level        | -          |     | $DS+(WS-DS)*n(m-1)$    |     |       | 8-3   |
| T update | Image update time  | at 25 °C   |     | 3                      | -   | sec   |       |
| Life     |                    | Topr       |     | 1000000times or 5years |     |       |       |

### Notes:

8-1. Luminance meter: Eye-One Pro Spectrophotometer.

8-2. CR=Surface Reflectance with all white pixel/Surface Reflectance with all black pixels.

8-3 WS: White state, DS: Dark state

## 9. Handling, Safety and Environment Requirements

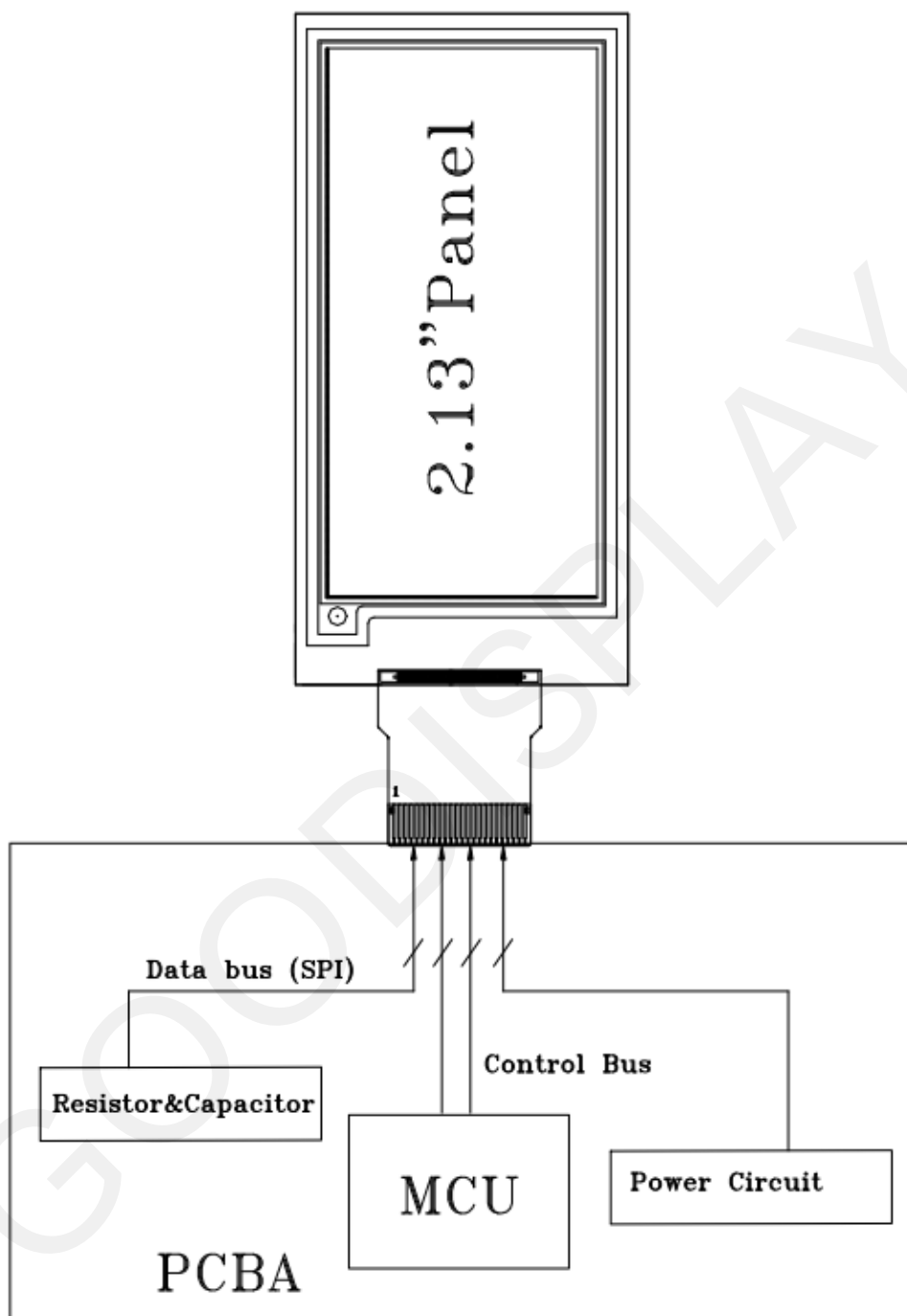
| <b>Warning</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| <p>The display glass may break when it is dropped or bumped on a hard surface. Handle with care. Should the display break, do not touch the electrophoretic material. In case of contact with electrophoretic material, wash with water and soap.</p>                                                                                                                                                                                                                                                                                  |                                                        |
| <b>Caution</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                        |
| <p>The display module should not be exposed to harmful gases, such as aalkaligases,whichcorrodeelectronic components. Disassembling the display module.</p> <p>Disassembling the display module can cause permanent damage and invalidates the warranty agreements.</p> <p>Observe general precautions that are common to handling delicate electronic components. The glass can break and front surfaces can easily be damaged. Moreover the display is sensitive to static electricity and other rough environmental conditions.</p> |                                                        |
| <b>Data sheet status</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                        |
| Product specification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | This data sheet contains final product specifications. |
| <b>Limiting values</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                        |
| <p>Limiting values given are in accordance with the Absolute Maximum Rating System (IEC 134).Stress above one or more of the limiting values may cause permanent damage to the device. These are stress ratings only and operation of the device at these or at any other conditions above those given in the Characteristics sections of the specification is not implied. Exposure to limiting values for extended periods may affect device reliability.</p>                                                                        |                                                        |
| <b>Application information</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                        |
| <p>Where application information is given, it is advisory and does not form part of the specification.</p>                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                        |

## 10. Reliability test

| NO | Test items                                | Test condition                                                                                                                                                                                                                              |
|----|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Low-Temperature Storage                   | T = -25°C, 240 h<br>Test in white pattern                                                                                                                                                                                                   |
| 2  | High-Temperature Storage                  | T=70° C, RH=40%, 240h<br>Test in white pattern                                                                                                                                                                                              |
| 3  | High-Temperature Operation                | T=50° C, RH=35%, 240h                                                                                                                                                                                                                       |
| 4  | Low-Temperature Operation                 | 0° C, 240h                                                                                                                                                                                                                                  |
| 5  | High-Temperature, High-Humidity Operation | T=40° C, RH=80%, 240h                                                                                                                                                                                                                       |
| 6  | High Temperature, High Humidity Storage   | T=50° C, RH=80%, 240h<br>Test in white pattern                                                                                                                                                                                              |
| 7  | Temperature Cycle                         | 1 cycle:[-25° C 30min]→[+70 ° C 30 min] : 50 cycles<br>Test in white pattern                                                                                                                                                                |
| 8  | UV exposure Resistance                    | 765W/m <sup>2</sup> for 168hrs, 40 °C<br>Test in white pattern                                                                                                                                                                              |
| 9  | ESD Gun                                   | Air+/-15KV; Contact+/-8KV<br>(Test finished product shell, not display only)<br>Air+/-8KV; Contact+/-6KV<br>(Naked EPD display, no including IC and FPC area)<br>Air+/-4KV; Contact+/-2KV<br>(Naked EPD display, including IC and FPC area) |

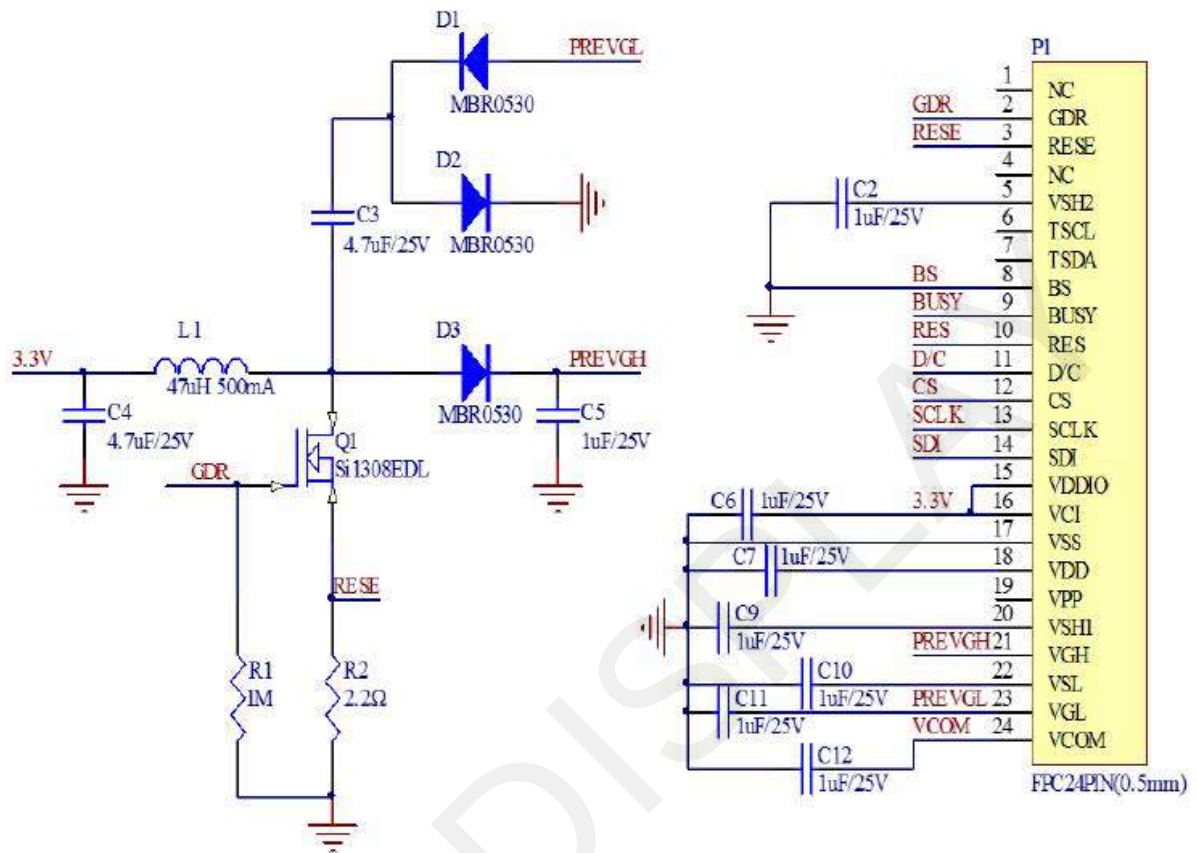
**Note:** Put in normal temperature for 1hour after test finished, display performance is ok.

## 11. Block Diagram





## 12. Reference Circuit



| Part Name | Requirements for spare part                                                                                       |
|-----------|-------------------------------------------------------------------------------------------------------------------|
| C1—C12    | 0603/0805; X5R/X7R; Voltage Rating: $\geq 25V$                                                                    |
| R1、R2     | 0603/0805; 1% variation, $\geq 0.05W$                                                                             |
| D1—D3     | MBR0530: 1) Reverse DC Voltage $\geq 30V$ 2) $I_o \geq 500mA$ 3) Forward voltage $\leq 430mV$                     |
| Q1        | Si1308EDL: 1) Drain-Source breakdown voltage $\geq 30V$ 2) $V_{gs(th)} \leq 1.5V$ 3) $R_{ds(on)} \leq 400m\Omega$ |
| L1        | refer to NR3015: $I_o = 500mA(max)$                                                                               |
| P1        | 24pins, 0.5mm pitch                                                                                               |

### **13. Matched Development Kit**

Our Development Kit designed for SPI E-paper Display aims to help users to learn how to use E-paper Display more easily. It can refresh black-white E-paper Display and three-color (black, white and red/Yellow) Good Display `s E-paper Display. And it is also added the functions of USB serial port, Raspberry Pi and LED indicator light ect.

DESPI Development Kit consists of the development board and the pinboard.

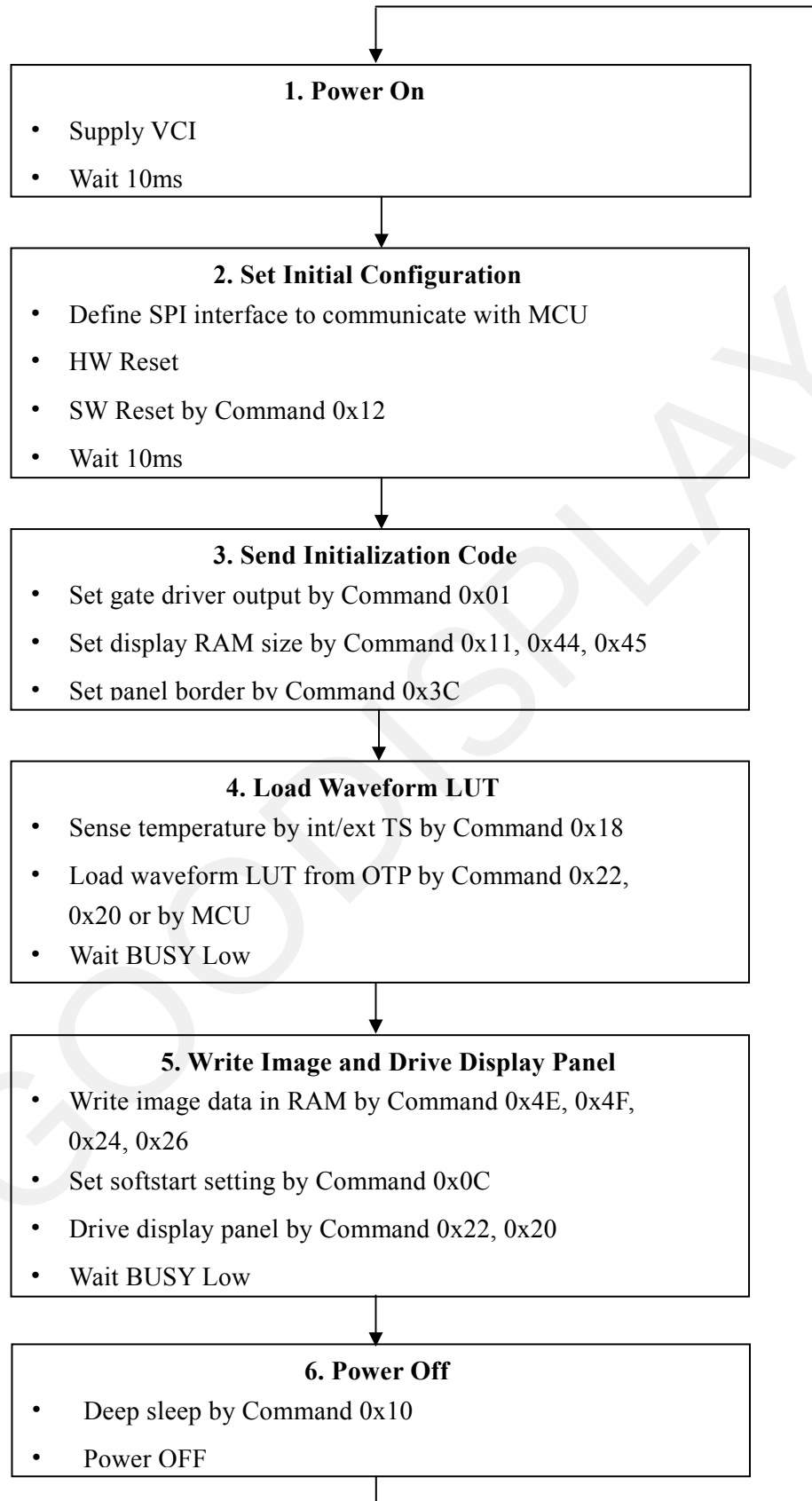
More details about the Development Kit, please click to the following link:

<https://www.good-display.com/product/53/>

GOODDISPLAY

## 14. Typical Operating Sequence

### 14.1 Normal Operation Flow



## 15. Inspection condition

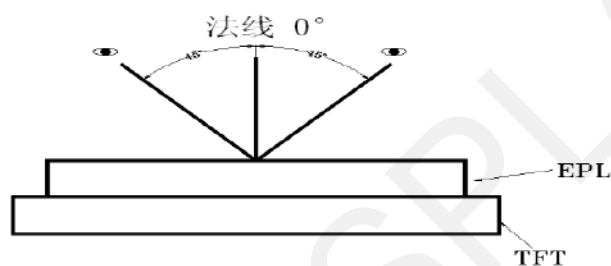
### 15. 1 Environment

Temperature:  $25\pm 3^{\circ}\text{C}$  Humidity:  $55\pm 10\%\text{RH}$

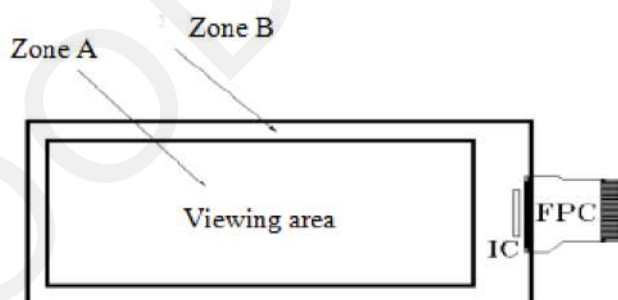
### 15. 2 Illuminance

Brightness: 1200~1500LUX; distance: 20-30CM; Angle: Relate  $30^{\circ}$  surround.

### 15.3 Inspection method



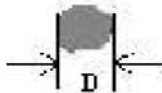
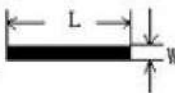
### 15. 4 Display area



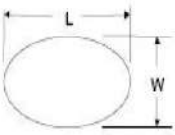
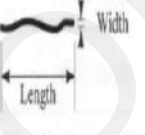


## 15. 5 Inspection standard

### 15. 5.1 Electric inspection standard

| NO. | Item                                                 | Standard                                                                                                                                                                                                                                        | Defect level | Method                     | Scope            |
|-----|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------------------------|------------------|
| 1   | Display                                              | Display complete<br>Display uniform                                                                                                                                                                                                             | MA           | Visual inspection          | Zone A           |
| 2   | Black/White spots                                    | <br>$D \leq 0.25\text{mm}$ , Allowed<br>$0.25\text{mm} < D \leq 0.4\text{mm}$ 。 $N \leq 3$ , and<br>Distance $\geq 5\text{mm}$<br>$0.4\text{mm} < D$ Not Allow | MI           |                            |                  |
| 3   | Black/White spots<br>(No switch)                     | <br>$L \leq 0.6\text{mm}$ , $W \leq 0.2\text{mm}$ , $N \leq 1$<br>$L \leq 2.0\text{mm}$ , $W > 0.2\text{mm}$ , Not Allow<br>$L > 0.6\text{mm}$ , Not Allow    |              |                            |                  |
| 4   | Ghost image                                          | Allowed in switching process                                                                                                                                                                                                                    | MI           | Visual inspection          |                  |
| 5   | Flash spots/<br>Larger FPL size                      | Flash spots in switching, Allowed<br>FPL size larger than viewing area,<br>Allowed                                                                                                                                                              | MI           | Visual/<br>Inspection card | Zone A<br>Zone B |
| 6   | Display<br>wrong/Missing                             | All appointed displays are showed<br>correct                                                                                                                                                                                                    | MA           | Visual<br>inspection       | Zone A           |
| 7   | Short circuit/<br>Circuit break/<br>Display abnormal | Not Allow                                                                                                                                                                                                                                       |              |                            |                  |

## 15.5.2 Appearance inspection standard

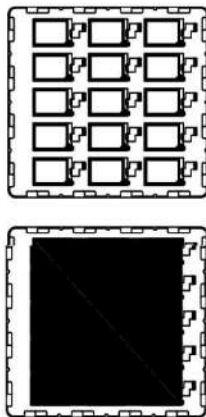
| NO. | Item                                              | Standard                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Defect level | Method                 | Scope              |
|-----|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|------------------------|--------------------|
| 1   | B/W spots<br>/Bubble/<br>Foreign bodies/<br>Dents |  $D = (L + W) / 2$ <p> <math>D \leq 0.25\text{mm}</math>, Allowed<br/> <math>0.25\text{mm} &lt; D \leq 0.4\text{mm}</math>, <math>N \leq 3</math><br/> <math>D &gt; 0.4\text{mm}</math>, Not Allow </p>                                                                                                                                                                                                                                                                                                                        | MI           | Visual inspection      | Zone A             |
| 2   | Glass crack                                       | Not Allow                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | MA           | Visual<br>/ Microscope | Zone A<br>Zone B   |
| 3   | Dirty                                             | Allowed if can be removed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | MI           |                        | Zone A<br>Zone B   |
| 4   | Chips/Scratch/<br>Edge crown                      |  <p> <math>X \leq 3\text{mm}</math>, <math>Y \leq 0.5\text{mm}</math> And without affecting the electrode is permissible </p>  <p> <math>2\text{mm} \leq X</math> or <math>2\text{mm} \leq Y</math> Not Allow </p>  <p> <math>W \leq 0.1\text{mm}</math>, <math>L \leq 5\text{mm}</math>, No harm to the electrodes and <math>N \leq 2</math> allow </p> | MI           | Visual<br>/ Microscope | Zone A<br>Zone B   |
| 5   | TFT Cracks                                        |  <p>Not Allow</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | MA           | Visual<br>/ Microscope | Zone A<br>Zone B   |
| 6   | Dirty/ foreign body                               | Allowed if can be removed/ allow                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | MI           | Visual<br>/ Microscope | Zone A /<br>Zone B |
| 7   | FPC broken/<br>Goldfingers<br>oxidation/ scratch  |   <p>Not Allow</p>                                                                                                                                                                                                                                                                                                                                                                                                                        | MA           | Visual<br>/ Microscope | Zone B             |

|    |                                                           |                                                                                                                                                                                                                                                       |    |                        |                  |
|----|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|------------------------|------------------|
| 8  | TFT edge bulge<br>/TFT chromatic<br>aberration            | TFT edge bulge:<br>$X \leq 3\text{mm}$ , $Y \leq 0.3\text{mm}$ Allowed<br>TFT chromatic aberration :Allowed                                                                                                                                           | MI | Visual<br>/ Microscope | Zone A<br>Zone B |
| 9  | PCB damaged/<br>Poor welding/<br>Curl                     | PCB (Circuit area) damaged Not<br>Allow<br>PCB Poor welding Not Allow<br>PCB Curl $\leq 1\%$                                                                                                                                                          | MI | Visual<br>/ Ruler      | Zone B           |
| 10 | Edge glue height/<br>Edge glue bubble                     | Edge Adhesives $H \leq$ PS surface<br>(Including protect film) Edge<br>adhesives seep in $\leq 1/2$ Margin width<br>Length excluding<br>Edge adhesives bubble: bubble<br>Width<br>$\leq 1/2$ Margin width; Length<br>$\leq 0.5\text{mm}$ 。 $n \leq 5$ |    |                        |                  |
| 11 | Protect film                                              | Surface scratch but not effect protect<br>function, Allowed                                                                                                                                                                                           |    | Visual<br>Inspection   |                  |
| 12 | Silicon glue                                              | Thickness $\leq$ PS surface(With protect film);<br>Full cover the IC;<br>Shape:<br>The width on the FPC $\leq 0.5\text{mm}$ (Front)<br>The width on the FPC $\leq 1.0\text{mm}$ (Back)<br>smooth surface, No obvious raised.                          | MI | Visual<br>Inspection   |                  |
| 13 | Warp degree<br>(TFT substrate)                            | $t \leq 2.0\text{mm}$                                                                                                                                                                                                                                 | MI | Ruler                  |                  |
| 14 | Color difference<br>in COM area<br>(Silver point<br>area) | Allowed                                                                                                                                                                                                                                               |    | Visual<br>Inspection   |                  |

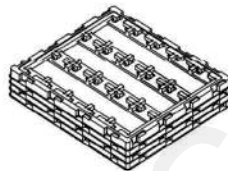
## 16. Packing

### PACKLING ORDER:

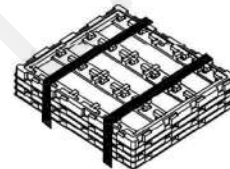
1) Putting 28 pcs Modules on each PET tray. And cover a dedicated EPE film.



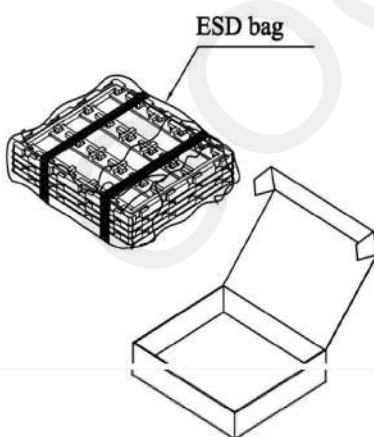
2) Putting 9 pcs PET trays together with 1 empty tray on the top of PET tray.



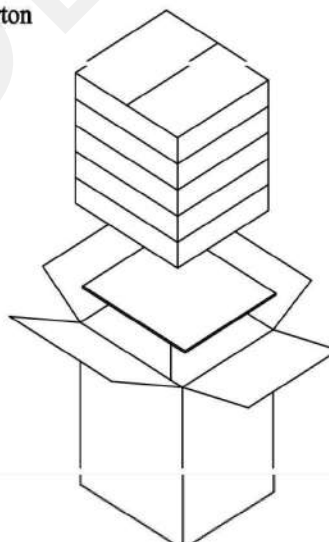
3) the tray together with rubber band



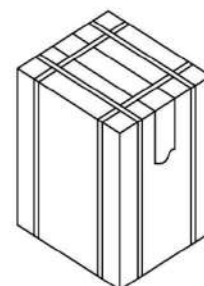
4) Insert in the ESD bag, add desiccant in the ESD bag, Putting in the inner small carton (TYPE:H82)



5) Putting 5 small cartons into one outcarton



6) Packing finished



Note: 28 pcs in a tray, 9 trays in a inner carton, 5 inner cartons in a out carton, so  $28 \times (10-1) \times 5 = 1260 \text{ pcs/Outcarton}$

Dimension (Small carton ): 385\*325\*87mm

Dimension (Out carton ): 394\*344\*470mm

## 17. Precautions

- (1) Do not apply pressure to the EPD panel in order to prevent damaging it.
- (2) Do not connect or disconnect the interface connector while the EPD panel is in operation.
- (3) Do not touch IC bonding area. It may scratch TFT lead or damage IC function.
- (4) Please be mindful of moisture to avoid its penetration into the EPD panel, which may cause damage during operation.
- (5) High temperature, high humidity, sunlight or fluorescent light may degrade the EPD panel's performance. Please do not expose the unprotected EPD panel to high temperature, high humidity, sunlight, or fluorescent for long periods of time.
- (6) For more precautions, please click on the link:

<https://www.good-display.com/news/80.html>